

What Should a Parametric PINN Be Conditioned On?

Encoding Coefficients, Initial Conditions and Boundaries Jointly, and What the Representation Does and Does Not Buy

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Abstract

Parametric physics-informed surrogates amortise the cost of solving a PDE family by conditioning one network on the PDE coefficients. The initial and boundary conditions, however, enter only through the training loss, so they are frozen into the weights: such a model cannot be queried at a new initial profile at all. We replace the coefficient encoder of a parametric PINN with a *specification encoder* on the coefficients, the initial condition and the boundary condition, holding the decoder, the coordinate encoder, the composite loss and its weights, the point budgets, the optimiser, the epoch budget and the seeds fixed, and we separate two questions that are usually run together: whether the extra conditioning works, and whether its *representation* matters.

The first answer is a large yes. One model generalises to unseen initial-condition parameters, to two functional families of initial conditions held out of training (Gaussian bumps, including the reference benchmark’s own, and three-mode superpositions), to coefficient $\times$ profile combinations never seen together, to three boundary specifications — two genuinely distinct boundary-value problems — against a non-periodic reference solver we author and validate to 1.1×10^{-4} against an independent stiff integration, and across five PDE families at once — reaching 0.3055 relative L_2 on a held-out functional family where the coefficient-only surrogate, which is not a function of the profile, returns 0.5529. Shuffling or freezing the encoded profile at evaluation time collapses that accuracy, so the conditioning is genuinely used. The capability is not free: at a fixed budget the conditioned model is up to $\times 2.47$ less accurate at the single specification the narrow model was built for, and we report that as a failed non-regression check.

The second answer is no, and we consider it the more useful result. We designed a symbol-aligned modal encoder — level/scale/shape factorisation, a modal expansion in the eigenbasis *of the boundary condition*, per-mode tokens carrying the closed-form action of the operator’s linear part, amplitude-weighted permutation-invariant pooling — and compared it against a budget-matched control that simply appends the sampled profile, both tuned on the same seen-specification split. It beats that control significantly on one of four families and is level or behind on the rest — and that single win disappears when the *same* control is given twice as many sensors at the same parameter budget, so we withdraw it: the structured encoding does not *significantly* beat a well-configured sampled-value encoding on any axis of any family. Two mechanism hypotheses built to explain the apparent win are reported as refuted, one of them by a fifth experiment run specifically to test it. A retraining oracle shows why there was so little to win: the decoder, not the encoder, is the binding constraint at this budget — the structured code provably carries about twice as much about an unseen functional family, and it makes no difference to the error. The one thing structure does measurably is absorb a boundary encoding absent from training: a 22 % error reduction — though a latent diagnostic shows both encoders place the unseen boundary equally correctly, so even there the mechanism is only partly pinned down. We also map a generalisation boundary in initial-condition sharpness that survives $3\times$ encoder width, $3\times$ epochs and $2\times$ collocation against a converged reference, and report an honest negative on transfer to an unseen PDE type, with the mechanism.

1 Introduction

A parametric PDE is never solved once. Design optimisation, uncertainty quantification and inverse problems all require the same equation at many points of a *problem-specification space*: many coefficient vectors, but also many initial profiles and many boundary configurations. Training one physics-informed neural network [Raissi et al., 2019] per specification is prohibitive, which has motivated a line of amortised surrogates that fold the specification into the network.

Parametric PINNs of the P²INN type [Cho et al., 2024] amortise across *coefficients*. A parameter encoder maps the coefficient vector μ to a latent code, a coordinate encoder maps (x, t) to a second

code, and a decoder consumes their concatenation; the explicit encoding of μ (as opposed to appending it to the coordinates) is what makes the surrogate work in the regimes where per-instance PINNs fail. But the initial and boundary conditions never enter the encoder. They enter only the *loss* — the initial condition sets the profile that the data term matches, the boundary condition sets the boundary term — and are therefore frozen into the trained weights. Such a surrogate is a function of (x, t, μ) alone: it can be queried at a new coefficient, but a new initial profile or a new boundary type requires a new model.

This paper asks what happens when that restriction is lifted, and — because the answer turns out to have two parts — separates two questions that the literature usually runs together:

- Q1 (capability)** Can the parameter encoder be replaced by a *specification encoder* on (μ, u_0, b) so that one model, at the same budget, answers queries at unseen initial conditions, unseen *functional families* of initial conditions, unseen coefficient $\times$ profile combinations, several boundary types and several PDE families?
- Q2 (representation)** Does it matter *how* the initial condition is represented — a structured, physics-aligned encoding versus the standard branch-network answer of raw sampled values — once both are given the same capacity, budget and tuning?

What we find. The answer to Q1 is an unqualified yes, and the effect is large. The answer to Q2, after a control that we very nearly did not run, is *no* — and we think that negative is the more useful contribution, because it locates the bottleneck.

- 1. Conditioning works, and it is what matters.** A single specification-conditioned model reaches 0.3055 relative L_2 on an entire functional family of initial conditions held out of training, where the coefficient-only surrogate — the same decoder, the same budget — returns 0.5529, because it is not a function of the profile at all. A coefficient-only model trained on *every* profile but given no profile input, which can only learn the profile-averaged solution, reaches 0.4574. The gap is present on all four PDE families we study, on all five held-out axes, and it is many times the across-seed spread.
- 2. The conditioning is genuinely used.** Replacing the encoded profile at evaluation time by another specification’s profile, by a constant, or by the reference training profile — always leaving the coefficients correct — degrades held-out accuracy sharply (e.g. 0.3063 $\rightarrow$ 0.4224 $\rightarrow$ 0.4916 on the held-out Gaussian family). Between 62% and 85% of the latent’s variance is attributable to the initial condition.
- 3. The representation is largely not the deciding factor.** We designed a symbol-aligned modal encoder — a level/scale/shape split, a modal expansion in the eigenbasis of the *boundary condition*, per-mode tokens carrying the closed-form action of the linear part of the operator, amplitude-weighted permutation-invariant pooling — and compared it against a matched-capacity control that simply appends the sampled profile, with both tuned on the same seen-specification validation split and their encoders matched to 0.26 % in parameter count. Across four families the structured encoder is significantly better on one (the reaction family, $p \leq 0.045$ by a rank test over 15 seeds), better-but-not-significantly on one, tied on one and *behind* on one — and that single win disappears when the same control is given $M=64$ sensors instead of 32 at the same parameter budget. We therefore withdraw it. A mechanism hypothesis we formed from the first three families made a falsifiable prediction; a fourth family run to test it refuted that too. On our own pre-registered noise criterion, taken literally, nothing separates anywhere.
- 4. The one thing structure does measurably: absorbing an unseen boundary encoding.** A model trained on periodic and Dirichlet data only, then queried under Neumann data — an unseen grid, transform, boundary term and one-hot component — reaches 0.2158 with the structured encoder against 0.2751 for the control — the only measured effect of the representation left standing. A latent diagnostic then narrows what can be claimed for it: both encoders place the unseen boundary equally correctly, so the advantage is in what the code contains, not where it sits. We also report the measurement that limits how much this shows: on the profiles Neumann compatibility forces, the Neumann and periodic *solutions* coincide, so three boundary specifications are only two distinct boundary-value problems.

5. **Why the representation matters so little elsewhere: the decoder is the bottleneck.** A coefficient-only model *retrained from scratch on a held-out profile* — the very cost the surrogate exists to avoid — is only 0.007 to 0.127 relative L_2 better than querying the conditioned model zero-shot, and on the hardest profile it is still at 0.6237. At this budget a 50-wide residual decoder, not the specification code, sets the accuracy, so there is little headroom for the encoding to express itself; three further diagnostics point the same way.
6. **Breadth and an honest negative.** One model supports 3 boundary specifications against a non-periodic reference solver we author and validate to 1.1×10^{-4} against an independent stiff integration, and one model supports 5 PDE families with transfer to held-out coefficient $\times$ profile compositions. Transfer to a PDE *type* absent from training in every form fails, under two documented attempts, and we explain why through the representation.

Everything is done at the reference parametric-PINN configuration: the coordinate encoder, the residual decoder with its hyper-shift, the concatenation injection, the composite loss and its equal weights, the point budgets, the optimiser, the epoch budget and the seeds are all held fixed, and the specification reaches the decoder *only* through the encoder output.

2 Setting

Equations. We work with the one-dimensional convection–diffusion–reaction family on the periodic interval $x \in [0, 2\pi)$, $t \in [0, 1]$,

$$u_t + \beta u_x - \nu u_{xx} - \rho u(1 - u) = 0, \quad u(x, 0) = u_0(x), \quad (1)$$

with $\mu = (\beta, \nu, \rho)$ and the Fisher reaction term. Setting individual coefficients to zero yields convection, diffusion, reaction, convection–diffusion and reaction–diffusion as special cases. The reaction term is what makes the solution a *nonlinear* functional of u_0 : for $\nu = 0$ the solution is the pointwise logistic map $u(x, t) = u_0 e^{\rho t} / (1 - u_0 + u_0 e^{\rho t})$, and for $\nu > 0$ the reaction and diffusion half-steps do not commute. By contrast, for $\rho = 0$ the solution is a single *linear* operator applied to the initial condition, $u = \mathcal{F}^{-1}[\hat{u}_0 e^{(\nu \lambda_k - i\beta k)t}]$; on such families initial-condition generalisation is close to free and does not discriminate between conditioning designs. Every headline result below is therefore stated on a reaction-bearing family.

The surrogate. The reference parametric PINN factorises the solution network as

$$u_\Theta(x, t; \mu) = g_{\theta_g}([g_{\theta_c}(x, t); g_{\theta_p}(\mu)]), \quad (2)$$

with a coordinate encoder g_{θ_c} producing $\mathbf{h}_{\text{coord}} \in \mathbb{R}^{50}$, a parameter encoder g_{θ_p} producing $\mathbf{h}_{\text{spec}} \in \mathbb{R}^{50}$, and a residual decoder g_{θ_g} on their concatenation; a linear hyper-shift $\text{modvec}(\beta, \nu, \rho, x, t)$ is added inside the decoder. All three parts are trained jointly on the composite physics-informed objective

$$\mathcal{L} = \underbrace{\frac{1}{N_f} \sum \mathcal{F}(x, t, \hat{u}; \mu)^2}_{\text{residual}} + \underbrace{\frac{1}{N_u} \sum \mathcal{F}(x, 0, \hat{u}; \mu)^2}_{\text{residual at } t=0} + \underbrace{\frac{1}{N_u} \sum (\hat{u}(x, 0) - u_0(x))^2}_{\text{initial data}} + \underbrace{\mathcal{L}_b}_{\text{boundary}}, \quad (3)$$

with all four weights equal to one. Only ground truth at $t = 0$ enters training; reference solutions elsewhere are used exclusively for evaluation.

What is conditioned and what is frozen. In Eq. (2) the encoder input is the coefficient vector and nothing else. The initial condition appears only as the target u_0 of the third term of Eq. (3), and the boundary condition only through $\mathcal{L}_b$. Both are properties of the objective, fixed once training starts. Consequently a trained model is a function of (x, t, μ) alone: it can be queried at a new coefficient, but a new initial profile or a new boundary type requires a new model. Our object of study is the enlarged conditioning

$$u_\Theta(x, t; \sigma) = g_{\theta_g}([g_{\theta_c}(x, t); E_\Theta(\sigma)]), \quad \sigma = (\mu, u_0, b), \quad (4)$$

where σ is a full problem specification — coefficients, initial condition, boundary condition — and E_Θ is a *specification encoder* that replaces g_{θ_p} and outputs the same $\mathbf{h}_{\text{spec}} \in \mathbb{R}^{50}$ into the same concatenation. The loss terms of Eq. (3) keep their form and their equal weights; the initial-data target and the boundary term are simply evaluated for whatever specification is being trained. Nothing else about the network changes: in particular modvec keeps its coefficient-only input in every arm, so the initial and boundary conditions reach the decoder *only* through $\mathbf{h}_{\text{spec}}$ — a strictly weaker injection than the coefficient enjoys.

problem specification $\sigma = (\mu, u_0, b)$

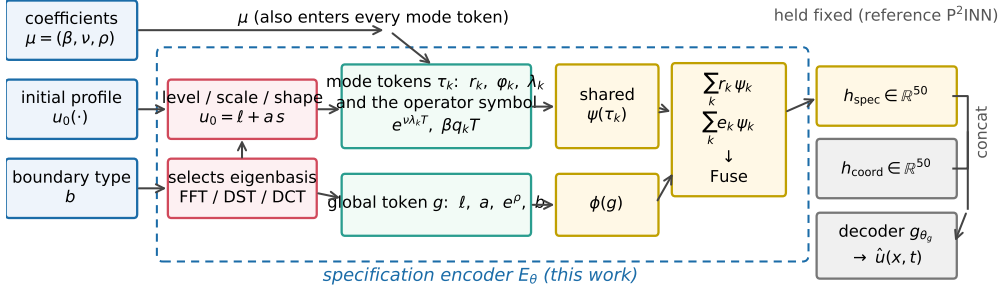


Figure 1: The specification encoder. Everything to the right of the dashed box is the reference parametric PINN, unchanged: the coordinate encoder, the concatenation injection and the residual decoder with its hyper-shift, whose input stays (β, ν, ρ, x, t) in every arm — so the initial and boundary conditions reach the decoder only through h_{spec} . Inside the box, the profile is split into level, scale and unit-RMS shape; the boundary condition selects the eigenbasis the shape is expanded in; each mode’s token carries its amplitude and phase together with the closed-form action of the operator’s linear part; one shared network processes every token and the tokens are pooled linearly in the modal amplitudes.

3 Symbol-aligned specification encoding

The design question is what function of $\sigma = (\mu, u_0, b)$ the encoder should see. A profile sampled at M sensors is the default answer in operator learning, and it is complete: no information is discarded. It is also, for our purposes, the worst-conditioned answer, because a network reading $\mathbb{R}^M$ sensor values must *learn* from a handful of training profiles the facts that (i) an overall rescaling of the profile acts very differently from a change of shape, (ii) the boundary condition determines which basis the profile should be read in, and (iii) the linear part of the operator acts diagonally on modes. Our encoder supplies all three.

3.1 Level, scale, shape

Write ℓ for the profile’s mean level ($\ell = u_0(0)$ under Dirichlet data, where the boundary level is the natural offset), $a = \|u_0 - \ell\|_{\text{RMS}}$ for its scale, and $s = (u_0 - \ell)/a$ for the unit-RMS shape, so $u_0 = \ell + a s$. This factorisation is not cosmetic. The reaction term $\rho u(1 - u)$ is a pointwise nonlinearity whose behaviour is governed almost entirely by the *level*: the logistic solution at the level, $\ell e^\rho / (1 - \ell + \ell e^\rho)$, already fixes the coarse temporal behaviour of the whole field. Diffusion, in contrast, acts on the shape and is scale-invariant. Splitting them lets (ℓ, a) and s be encoded on separate, each nearly one-dimensional, axes rather than being entangled in M correlated sensor values.

3.2 A boundary-adapted modal expansion

The shape is expanded in the eigenbasis of the Laplacian associated with the conditioned boundary condition:

boundary	grid	transform	λ_k
periodic	$x_j = jL/N$	real DFT	$-k^2$
Dirichlet	$x_j = jL/(N+1)$	DST-I	$-(k\pi/L)^2$
Neumann	$x_j = (j + \frac{1}{2})L/N$	DCT-II	$-(k\pi/L)^2$

giving coefficients $c_k = r_k e^{i\varphi_k}$, $k = 1, \dots, K$ (for the real transforms $\varphi_k \in \{0, \pi\}$). The boundary condition is therefore not merely a categorical label appended to the input: it *selects the coordinate system* in which the profile is described, and it changes the eigenvalues that the next step consumes. This is the same object the boundary term of the loss enforces, expressed on the input side.

3.3 Symbol tokens and amplitude-weighted pooling

For each retained mode we form a token (all entries scalars)

$$\tau_k = \left[\underbrace{\left[\frac{k}{K}, \frac{\lambda_k}{K^2}, r_k, \sqrt{e_k}, \cos \varphi_k, \sin \varphi_k, \rho, \nu, \frac{\nu \lambda_k T}{40} \right]}_{\text{mode } k \text{ of the shape}}, \underbrace{\left[e^{\nu \lambda_k T}, r_k e^{\nu \lambda_k T}, \cos \beta q_k T, \sin \beta q_k T, \frac{\beta q_k T}{1 + |\beta q_k T|} \right]}_{\text{PDE symbol on mode } k} \right] \quad (5)$$

where $e_k = r_k^2 / \sum_j r_j^2$ is the modal energy share, q_k the convective wavenumber ($q_k = k$ for periodic data, 0 otherwise) and $T = 1$ the horizon. The six bracketed entries are the closed-form action of the *linear* part of Eq. (1) on mode k : diffusion attenuates it by $e^{\nu \lambda_k T}$ and convection rotates its phase by $\beta q_k T$. They are not the solution — the reaction term, which is what makes the problem hard and the initial-condition dependence nonlinear, is absent — but they hand the encoder the one coefficient $\times$ profile interaction that is known analytically, instead of asking an MLP to discover νk^2 from a handful of examples.

A single shared two-layer network ψ processes every token, and the tokens are pooled by two permutation-invariant operators that are *linear in the modal amplitudes*,

$$\mathbf{h}_{\text{spec}} = \text{Fuse} \left(\underbrace{\sum_k r_k \psi(\tau_k)}_{\text{amplitude-weighted}}, \underbrace{\sum_k e_k \psi(\tau_k)}_{\text{energy-weighted}}, \phi(g) \right) \in \mathbb{R}^{50}, \quad (6)$$

where ϕ is a second two-layer network on a global token g holding $\ell, a, \log a, \ell(1 - \ell), e^\rho$, the logistic value at the level, $\min u_0, \max u_0, (\beta, \nu, \rho)$, the spectral centroid and bandwidth, and a boundary one-hot.

Weight sharing and amplitude-linear pooling are the two properties that carry generalisation across functional families. Sharing means the machinery that reads mode k is the machinery that read mode 1 during training, so energy appearing in a previously unexcited mode is not read by untrained weights. Linearity in r_k means that a profile which is a superposition contributes as a superposition: with ψ fixed, Eq. (6) is exactly the composition rule that the linear part of the operator obeys, so the encoder extrapolates along the one direction — modal amplitude — in which the training set is most degenerate. A single sinusoid, normalised to unit RMS, puts *all* its energy in one mode at one amplitude; a Gaussian bump spreads it over many modes at many amplitudes. Without amplitude weighting, an unweighted sum over K tokens is dominated by the $K - 1$ zero-amplitude tokens of the training profiles and does not transfer; we ablate exactly this in §5.3.

3.4 What is held fixed, and the control

The encoder output enters the *same* concatenation $[\mathbf{h}_{\text{spec}}; \mathbf{h}_{\text{coord}}] \in \mathbb{R}^{100}$ and the same decoder as in Eq. (2); the coordinate encoder, the decoder, the hyper-shift, the loss terms and their equal weights, the point budgets, the optimiser and the epoch budget are unchanged (Table 1).

The control we compare against is the standard branch-net design: the reference MLP encoder with μ , the profile sampled at M equispaced sensors and a boundary one-hot concatenated into its input. It receives the same information, the same injection surface, the same budget, and its own hyperparameters (M and width) tuned on the same seen-specification validation split. Its specification-encoder parameter count is matched to ours to within 0.26 %. It is the arm that isolates *structure*: any gain that merely comes from feeding the profile in at all is a gain the control also enjoys.

4 Experimental design

Every split, the decision rule and the hyperparameter protocol were fixed in a timestamped plan file (code/specification_encoder_plan.json, initial-condition fingerprint 07847503a7fc92ec) before any comparison was run. No held-out set is trained on, and none is used to select anything.

4.1 The specification space and its splits

PDE families and coefficients. Four core families, all special cases of Eq. (1): *reaction–diffusion* $\mu = (0, \nu, \rho)$ and *convection–reaction* $\mu = (\beta, 0, \rho)$, both on the 5×5 integer grid split by

parity of the coefficient sum into 13 trained and 12 held-out pairs; *reaction* $\mu = (0, 0, \rho)$ with $\rho \in \{1, 3, 5, 7, 9\}$ trained and $\rho \in \{2, 4, 6, 8, 10\}$ held out; and *convection–reaction at stiff* $\rho, \beta \in \{1, 3, 5\} \times \rho \in \{1, 3, 5, 7, 9\}$ trained, $\beta \in \{2, 4\} \times \rho \in \{2, 4, 6, 8\}$ held out. The first two are the pre-registered core; the last two were added during the study, the reaction family to obtain a non-linear initial-condition axis free of transport and diffusion, and the stiff convection–reaction family purely to test a hypothesis formed from the first three (§5.3). Both additions, and the fact that neither replaces a registered result, are recorded in the attempts log. For the breadth study (§8.2) five families — convection, diffusion, reaction, reaction–diffusion and CDR — are trained jointly.

Initial conditions. The reference implementation ships a zoo of initial conditions — amplitude-scaled sines, offset sines, a second frequency, two Gaussians of different width, and several other shapes. We read that zoo as three closed-form *families* of which its entries are special cases, and we train on exactly one of them:

- S** single-mode sinusoids $c + A \sin(kx + \varphi)$, $k \in \{1, 2\}$ (contains $\sin(x)$, $0.1 \sin(x)$, $0.5 \sin(x)$, $1 + \sin(x)$, $2 + \sin(x)$, $\sin(2x)$, $\sin^2(x)$, ...). **Trained:** 12 profiles, including the reference reaction–diffusion initial condition $1 + \sin(x)$. **Held out (parameters):** 8 further profiles of the same family at parameter values never trained on.
- G** Gaussian bumps $c + A \exp(-d(x, x_0)^2 / 2\sigma^2)$ (contains the zoo entries `gauss`, $\sigma = \pi/4$, and `gauss_pi_2`, $\sigma = \pi/2$). **Held out as a whole family:** 8 profiles, two of which are the zoo Gaussians generated by the *unmodified* reference construction.
- M** three-mode superpositions $c + \sum_{k \leq 3} a_k \sin(kx + \varphi_k)$; no zoo entry has this form. **Held out as a whole family:** 6 profiles.

All profiles are non-negative, because every reaction-bearing family integrates the logistic map, which blows up in finite time from negative data. Since no multi-mode and no localised profile is ever trained on, families **G** and **M** test genuine *functional* generalisation, not parameter interpolation.

Boundary conditions. Periodic (core), plus Dirichlet ($u = u_0(0)$ at both ends) and homogeneous Neumann for the boundary study (§8.1). The boundary term of Eq. (3) is the one belonging to the conditioned boundary type, with the same 100 abscissae per specification in all three cases.

Evaluation axes. *seen*: trained coefficients $\times$ trained profiles. *unseen coefficients*: held-out coefficients $\times$ trained profiles. *unseen IC parameters*: trained coefficients $\times$ family-S held-out profiles. *unseen IC family*: trained coefficients $\times$ family G or M. *unseen combination*: held-out coefficients $\times$ held-out profiles — neither seen, and never seen together. Errors are relative L_2 per specification over its 1000 test points, averaged over specifications, reported as mean $\pm$ standard deviation over 5 seeds (15 on the reaction family, whose training outcome is bimodal), together with a one-sided Mann–Whitney rank test over the per-seed errors.

4.2 Reference solutions and their validation

Periodic reference solutions come from the shipped construction, executed verbatim with the initial condition supplied as a new argument: the closed-form logistic solution for reaction, the spectral solution for convection/diffusion, and operator splitting for reaction–diffusion and CDR. Routing an initial condition through our bridge reproduces the shipped solver *bit for bit* (maximum absolute difference 0.0 over all tested family $\times$ zoo-profile pairs).

For non-periodic data we author a solver (§3 table): the same Lie–Trotter scheme — exact pointwise logistic reaction step, exact-in-time diffusion step — with the Laplacian diagonalised in the eigenbasis of the boundary condition, $n_{\text{sub}} = 64$ substeps per output step. Its validation record (Table 11) is: (i) run in the shipped time convention its periodic branch reproduces the shipped spectral reference to 4.2×10^{-15} in maximum norm; (ii) it converges under time and space refinement; (iii) it agrees with an *independent* stiff method-of-lines integration (`solve_ivp`, LSODA, $\text{rtol}=\text{atol}=10^{-10}$) to 1.1×10^{-4} relative L_2 in the worst case over all three boundary types. We additionally spot-check that the shipped operator-split reference is converged at the sharpest and highest-frequency profiles used anywhere in this study: doubling the spatial grid changes it by at most 1.5×10^{-7} , and halving the time step by at most 0.0047, both far below the decision threshold.

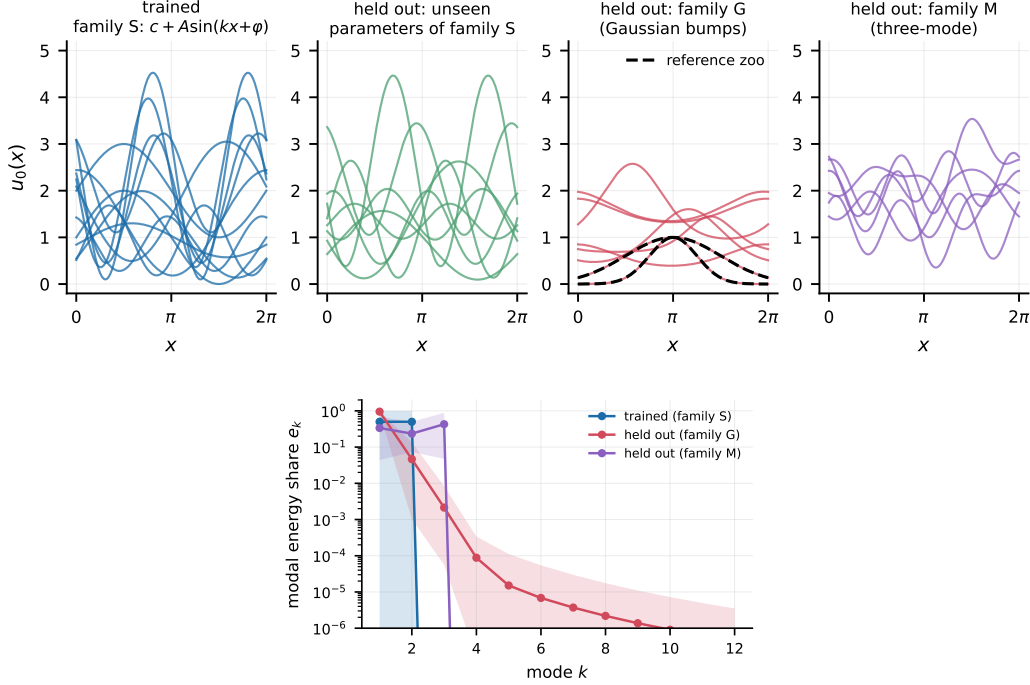


Figure 2: The initial-condition space. *Top:* the trained family (single-mode sinusoids), the held-out parameters of that family, and the two held-out functional families — Gaussian bumps, with the two zoo Gaussians produced by the unmodified reference construction dashed in black, and three-mode superpositions. *Bottom:* the modal energy of each set (mean, with the min–max band), which is what makes G and M functionally distinct rather than merely new parameter values: every trained profile puts all of its energy in a single mode $k \in \{1, 2\}$, family M spreads it over three, and family G decays smoothly out to $k \approx 10$.

4.3 Budget, protocol and decision rule

Hyperparameters. Every hyperparameter introduced by the conditioning — the number of retained modes K and the encoder width H for ours, the sensor count M and the width W for the control — was selected on the *seen-specification validation split*: the 1000 validation points per trained specification produced by the reference sampler, disjoint from both the collocation points and the test points. The grids were $K \in \{8, 12, 16\}$, $H \in \{64, 100, 128\}$, $M \in \{16, 32, 64\}$, $W \in \{150, 190\}$, two seeds each (30 runs, Figure 13); the selected settings are $K = 16$, $H = 100$, $M = 32$, $W = 150$. The resulting specification–encoder parameter counts agree to 0.26 %.

Decision rule (pre-registered). An improvement counts on an axis when it clears $\delta(x) = \max(0.01, 0.10x)$, with x the comparator’s mean relative L_2 , and exceeds the pooled across-seed standard deviation of the two arms. On the original coefficient-only setting, the conditioned model may not exceed the coefficient-only baseline by more than $\max(10\%, 0.005)$.

5 Joint coefficient $\times$ initial-condition generalisation

Reading the baselines. Two coefficient-only references frame the comparison. The first is the reference parametric PINN trained with the single fixed initial condition $1 + \sin(x)$; it is not a function of the profile at all, so on a held-out profile it can only return its trained solution. We report it at its own trained initial condition, which is the accuracy it can actually *reach* without retraining, and separately note what happens when it is queried at a new profile anyway. The second is the same coefficient-only encoder trained on *all* 156 specifications: it sees every profile in the loss but has no input that distinguishes them, so it can only learn the profile-averaged solution. It is the control for “does the model need to be conditioned on the initial condition at all”.

Table 1: Held-fixed configuration and disclosed capacities. The specification encoder is the only component that differs between arms.

component	setting	varies by arm?
coordinate encoder g_{θ_c}	$2 \rightarrow 50 \rightarrow 50 \rightarrow 50$, tanh	no
decoder g_{θ_g}	$100 \rightarrow 50$, 5 residual blocks, $50 \rightarrow 1$	no
hyper-shift modvec	Linear(5, 50) on (β, ν, ρ, x, t)	no
injection	$[\mathbf{h}_{\text{spec}}; \mathbf{h}_{\text{coord}}] \in \mathbb{R}^{100}$, $\mathbf{h}_{\text{spec}} \in \mathbb{R}^{50}$	no
loss	$\mathcal{L}_f + \mathcal{L}_{f,0} + \mathcal{L}_u + \mathcal{L}_b$, weights 1	no
points / specification	1000 collocation, 256 initial, 100 boundary pairs	no
evaluation / specification	1000 test, 1000 validation (disjoint)	no
optimiser, epochs	Adam (default lr), 20 000	no
coefficient-only encoder	$3 \rightarrow 150^3 \rightarrow 50$	53,450 params
sampling-profile control	$(3+M+3) \rightarrow W^3 \rightarrow 50$, $M=32$, $W=150$	58,700 params
symbol-aligned encoder	$K=16$ modes, width $H=100$	58,550 params

Table 2: Relative L_2 (mean $\pm$ s.d. over 5 seeds; absolute error in the small row) on the seen specifications and on every held-out axis, at a matched budget. The coefficient-only baseline is shown twice: as it behaves when *queried* at a held-out specification, and — since it cannot in fact be queried at a new initial condition — at its own trained initial condition, the accuracy it can reach without retraining. Columns: seen specifications; held-out coefficient μ^* ; held-out parameters of the trained IC family u_0^* ; the two held-out *functional* families G (Gaussian bumps, including the benchmark’s own) and M (three-mode); and a held-out coefficient *and* profile, never seen together. The last column is the specification-encoder parameter count: the two conditioned arms are matched to 0.26 %, and everything else — decoder, coordinate encoder, loss, collocation, optimiser, epochs, seeds — is identical across all rows. Nothing is emphasised: the only axis on which the structured arm beats this control significantly is the reaction family, and §5.3 shows that advantage does not survive against a denser-sensor configuration of the *same* control at the same budget.

family	method	seen	μ^*	u_0^*	fam. G	fam. M	$\mu^* \times u_0^*$	enc. par.
conv.-reac. $\rho \leq 5$	coeff.-only P ² INN, <i>queried</i> at the new spec.	0.4137 $\pm 0.003/0.346$	0.4120 $\pm 0.005/0.341$	0.4408 $\pm 0.003/0.363$	0.5529 $\pm 0.004/0.361$	0.4480 $\pm 0.004/0.373$	0.4772 $\pm 0.008/0.358$	53,450
	<i>same model at its own trained IC</i>	0.0699 $\pm 0.007/0.026$	0.1010 $\pm 0.020/0.053$	0.0699 $\pm 0.007/0.026$	0.0699 $\pm 0.007/0.026$	0.0699 $\pm 0.007/0.026$	0.1010 $\pm 0.020/0.053$	53,450
	coeff.-only, all ICs in the loss, no IC input	0.3192 $\pm 0.001/0.229$	0.3169 $\pm 0.001/0.225$	0.3027 $\pm 0.001/0.227$	0.4574 $\pm 0.003/0.273$	0.1866 $\pm 0.001/0.143$	0.3237 $\pm 0.001/0.217$	53,450
	sampling-profile control	0.0863 $\pm 0.037/0.066$	0.0832 $\pm 0.028/0.064$	0.2309 $\pm 0.047/0.192$	0.3257 $\pm 0.043/0.197$	0.1923 $\pm 0.014/0.156$	0.2530 $\pm 0.035/0.181$	58,700
	symbol-aligned encoder (ours)	0.0696 $\pm 0.002/0.049$	0.0759 $\pm 0.002/0.057$	0.1989 $\pm 0.014/0.160$	0.3055 $\pm 0.013/0.187$	0.1865 $\pm 0.022/0.151$	0.2328 $\pm 0.014/0.165$	58,550
reac.-diff. $\rho \leq 5$	coeff.-only P ² INN, <i>queried</i> at the new spec.	0.3263 $\pm 0.011/0.269$	0.3247 $\pm 0.012/0.264$	0.3451 $\pm 0.011/0.279$	0.4221 $\pm 0.011/0.263$	0.3770 $\pm 0.008/0.327$	0.3786 $\pm 0.011/0.281$	53,450
	<i>same model at its own trained IC</i>	0.1005 $\pm 0.020/0.074$	0.1026 $\pm 0.021/0.076$	0.1005 $\pm 0.020/0.074$	0.1005 $\pm 0.020/0.074$	0.1005 $\pm 0.020/0.074$	0.1026 $\pm 0.021/0.076$	53,450
	coeff.-only, all ICs in the loss, no IC input	0.2319 $\pm 0.002/0.158$	0.2291 $\pm 0.002/0.153$	0.2120 $\pm 0.002/0.147$	0.4025 $\pm 0.004/0.235$	0.1536 $\pm 0.002/0.122$	0.2613 $\pm 0.002/0.168$	53,450
	sampling-profile control	0.1286 $\pm 0.007/0.113$	0.1299 $\pm 0.007/0.113$	0.1857 $\pm 0.033/0.155$	0.2513 $\pm 0.022/0.155$	0.1802 $\pm 0.003/0.159$	0.2058 $\pm 0.011/0.154$	58,700
	symbol-aligned encoder (ours)	0.1225 $\pm 0.006/0.108$	0.1242 $\pm 0.006/0.109$	0.1800 $\pm 0.011/0.152$	0.2585 $\pm 0.012/0.161$	0.1728 $\pm 0.010/0.151$	0.2039 $\pm 0.006/0.152$	58,550
reaction $\rho \leq 9$	coeff.-only P ² INN, <i>queried</i> at the new spec.	0.4652 $\pm 0.199/0.399$	0.4548 $\pm 0.238/0.371$	0.4803 $\pm 0.200/0.409$	0.5551 $\pm 0.194/0.397$	0.4860 $\pm 0.200/0.424$	0.4855 $\pm 0.233/0.380$	53,450
	<i>same model at its own trained IC</i>	0.2480 $\pm 0.239/0.181$	0.2921 $\pm 0.289/0.217$	0.2480 $\pm 0.239/0.181$	0.2480 $\pm 0.239/0.181$	0.2480 $\pm 0.239/0.181$	0.2921 $\pm 0.289/0.217$	53,450
	coeff.-only, all ICs in the loss, no IC input	0.5830 $\pm 0.266/0.543$	0.6119 $\pm 0.324/0.571$	0.5666 $\pm 0.276/0.539$	0.6300 $\pm 0.298/0.517$	0.5258 $\pm 0.263/0.535$	0.6068 $\pm 0.339/0.564$	53,450
	sampling-profile control	0.3025 $\pm 0.198/0.286$	0.3678 $\pm 0.240/0.347$	0.4809 $\pm 0.217/0.459$	0.5754 $\pm 0.211/0.477$	0.4267 $\pm 0.236/0.424$	0.5223 $\pm 0.245/0.479$	58,700
	symbol-aligned encoder (ours)	0.1967 $\pm 0.254/0.179$	0.2450 $\pm 0.313/0.224$	0.3204 $\pm 0.233/0.281$	0.4080 $\pm 0.218/0.300$	0.3091 $\pm 0.226/0.280$	0.3524 $\pm 0.275/0.291$	58,550
conv.-reac. $\rho \leq 9$	coeff.-only P ² INN, <i>queried</i> at the new spec.	0.5974 $\pm 0.221/0.558$	0.6396 $\pm 0.263/0.593$	0.6130 $\pm 0.217/0.571$	0.6814 $\pm 0.203/0.543$	0.6179 $\pm 0.211/0.596$	0.6683 $\pm 0.249/0.600$	53,450
	<i>same model at its own trained IC</i>	0.3997 $\pm 0.284/0.332$	0.5035 $\pm 0.348/0.424$	0.3997 $\pm 0.284/0.332$	0.3997 $\pm 0.284/0.332$	0.3997 $\pm 0.284/0.332$	0.5035 $\pm 0.348/0.424$	53,450
	sampling-profile control	0.1494 $\pm 0.147/0.124$	0.1441 $\pm 0.149/0.119$	0.2439 $\pm 0.105/0.198$	0.3356 $\pm 0.143/0.224$	0.1966 $\pm 0.086/0.161$	0.2499 $\pm 0.100/0.179$	58,700
	symbol-aligned encoder (ours)	0.2379 $\pm 0.223/0.214$	0.2634 $\pm 0.241/0.238$	0.3348 $\pm 0.226/0.297$	0.4148 $\pm 0.226/0.311$	0.2954 $\pm 0.198/0.270$	0.3344 $\pm 0.208/0.272$	58,550

Table 3: Representation ablations at a matched specification-encoder budget, and the conditioning-removed evaluation. Every arm is trained with the identical decoder, loss, budget and seeds.

variant	seen	unseen IC par.	unseen fam. G	unseen fam. M	unseen comb.
symbol-aligned (ours)	0.0696 $\pm$ 0.0017	0.1989 $\pm$ 0.0141	0.3055 $\pm$ 0.0135	0.1865 $\pm$ 0.0223	0.2328 $\pm$ 0.0142
— unweighted pooling	0.1106 $\pm$ 0.0694	0.2351 $\pm$ 0.0177	0.3744 $\pm$ 0.0243	0.1881 $\pm$ 0.0160	0.2709 $\pm$ 0.0076
— no symbol features	0.0841 $\pm$ 0.0183	0.2088 $\pm$ 0.0137	0.3032 $\pm$ 0.0117	0.1968 $\pm$ 0.0018	0.2371 $\pm$ 0.0076
— flat MLP on same features	0.0708 $\pm$ 0.0034	0.2367 $\pm$ 0.0035	0.3587 $\pm$ 0.0054	0.2311 $\pm$ 0.0033	0.2795 $\pm$ 0.0026
sampled profile, M=32	0.0863 $\pm$ 0.0373	0.2309 $\pm$ 0.0472	0.3257 $\pm$ 0.0431	0.1923 $\pm$ 0.0139	0.2530 $\pm$ 0.0349
sampled profile, M=64	0.0783 $\pm$ 0.0067	0.2407 $\pm$ 0.0151	0.3394 $\pm$ 0.0117	0.1988 $\pm$ 0.0066	0.2636 $\pm$ 0.0012
<i>conditioning removed at evaluation (symbol-aligned encoder, coefficients kept correct)</i>					
true profile	0.0702 $\pm$ 0.0010	0.2039 $\pm$ 0.0095	0.3063 $\pm$ 0.0054	0.1763 $\pm$ 0.0054	0.2320 $\pm$ 0.0060
shuffled profile	0.3951 $\pm$ 0.0063	0.3474 $\pm$ 0.0113	0.4224 $\pm$ 0.0302	0.2200 $\pm$ 0.0047	0.3687 $\pm$ 0.0043
constant profile	0.4341 $\pm$ 0.0031	0.3656 $\pm$ 0.0065	0.4932 $\pm$ 0.0584	0.2047 $\pm$ 0.0360	0.3523 $\pm$ 0.0083
frozen to $1+\sin(x)$	0.3738 $\pm$ 0.0023	0.3907 $\pm$ 0.0031	0.4916 $\pm$ 0.0028	0.3880 $\pm$ 0.0041	0.4260 $\pm$ 0.0032

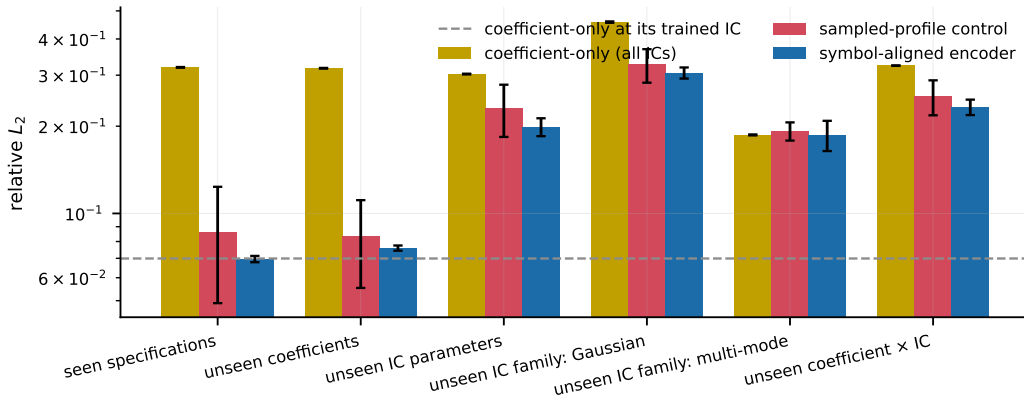


Figure 3: Convection–reaction. Mean relative L_2 on each evaluation axis (log scale, error bars are across-seed s.d.). The dashed line is what the coefficient-only surrogate reaches at its own trained initial condition — the best it can do without retraining, since it cannot be queried elsewhere.

5.1 Conditioning is what matters

Table 2 and Figure 3 answer Q1. On every family and every held-out axis, both conditioned arms are far more accurate than either coefficient-only reference, and the gap dwarfs the across-seed spread. On the convection–reaction family the structured encoder reaches 0.3055 on the held-out Gaussian family and 0.1865 on the held-out three-mode family, against 0.5529 and 0.4480 for the coefficient-only surrogate queried there — a factor of 1.81 and 2.40. The same holds on the unseen coefficient $\times$ profile combinations (0.2328 against 0.4772).

The comparison that isolates *conditioning* from *training data* is the third row: a coefficient-only encoder trained on all 156 specifications, so it sees every profile in the loss but has no input that distinguishes them. It can only learn the profile-averaged solution and reaches 0.4574 (0.4025 on reaction–diffusion) — better than the fixed-IC baseline, because it has at least seen the profile distribution, but far behind either conditioned model. Feeding the profile to the encoder, not merely putting it in the loss, is what buys the generalisation.

What it costs. This capability is not free at a fixed budget. On the *original* coefficient-only setting — the reference profile $1+\sin(x)$ at the trained coefficients — a model that also has to cover 12 profiles is less accurate than one trained on that profile alone: 0.1726 against 0.0699 on convection–reaction ($\times 2.47$), 0.1321 against 0.1005 on reaction–diffusion ($\times 1.32$), 0.2917 against 0.2480 on reaction ($\times 1.18$). Only 18/46 of the per-coefficient checks fall inside the tolerance we pre-registered ($\max(10\%, 0.005)$), so we record this as a *failed* non-regression check rather than a passed one (Table 12). The trade is explicit: one model covering a $12\times$ larger specification set, at the cost of up to $\times 2.47$ at the single specification the narrow model was built for. Both conditioned arms pay it equally, so it does not affect any comparison between them.

Table 4: Structured encoder against the matched control, per family and axis. “margin” is control minus ours; δ is the pre-registered threshold $\max(0.01, 0.1 \times \text{control})$; s_{pool} is the pooled across-seed standard deviation, the pre-registered noise criterion; p is a one-sided Mann–Whitney rank test over the per-seed errors, which does not assume the unimodality that s_{pool} implicitly does. Only the reaction family separates, and only under the rank test.

family	axis	ours	control	margin	δ / s_{pool}	p
conv.-reac. $\rho \leq 5$	seen	0.0696	0.0863	+0.0167	0.010 / 0.026	0.345
	u_0^*	0.1989	0.2309	+0.0320	0.023 / 0.035	0.075
	fam. G	0.3055	0.3257	+0.0202	0.033 / 0.032	0.345
	fam. M	0.1865	0.1923	+0.0058	0.019 / 0.019	0.210
	$\mu^* \times u_0^*$	0.2328	0.2530	+0.0202	0.025 / 0.027	0.155
reac.-diff. $\rho \leq 5$	seen	0.1225	0.1286	+0.0061	0.013 / 0.007	0.111
	u_0^*	0.1800	0.1857	+0.0056	0.019 / 0.025	0.579
	fam. G	0.2585	0.2513	-0.0072	0.025 / 0.018	0.726
	fam. M	0.1728	0.1802	+0.0074	0.018 / 0.007	0.111
	$\mu^* \times u_0^*$	0.2039	0.2058	+0.0019	0.021 / 0.009	0.274
reaction $\rho \leq 9$	seen	0.1967	0.3025	+0.1058	0.030 / 0.228	0.267
	u_0^*	0.3204	0.4809	+0.1605	0.048 / 0.225	0.031*
	fam. G	0.4080	0.5754	+0.1674	0.058 / 0.214	0.013*
	fam. M	0.3091	0.4267	+0.1176	0.043 / 0.231	0.045*
	$\mu^* \times u_0^*$	0.3524	0.5223	+0.1699	0.052 / 0.260	0.037*
conv.-reac. $\rho \leq 9$	seen	0.2379	0.1494	-0.0885	0.015 / 0.189	0.655
	u_0^*	0.3348	0.2439	-0.0910	0.024 / 0.176	0.500
	fam. G	0.4148	0.3356	-0.0792	0.034 / 0.189	0.421
	fam. M	0.2954	0.1966	-0.0988	0.020 / 0.153	0.889
	$\mu^* \times u_0^*$	0.3344	0.2499	-0.0846	0.025 / 0.163	0.579

5.2 The conditioning is genuinely used

Table 14 removes the conditioning at evaluation time while leaving the coefficients correct. The sharpest version of the test is on the *seen* specifications — the ones the model was trained on, where it has no excuse: feeding the encoder another specification’s profile raises the error from 0.0702 to 0.3951, a factor of 5.6; a single constant profile gives 0.4341 and freezing it at the reference training profile $1 + \sin(x)$ — which is exactly what the coefficient-only baseline does implicitly — gives 0.3738, i.e. back to coefficient-only behaviour. On the held-out Gaussian family the same substitutions give $0.3063 \rightarrow 0.4224$ (shuffled), 0.4932 (constant) and 0.4916 (frozen). The control behaves the same way ($0.0971 \rightarrow 0.4231$ on the seen specifications, a factor of 4.4), as it should: both models use the profile they are given. Between 62% and 85% of the latent’s variance across the trained specifications is attributable to the initial condition rather than to the coefficients, so the code is neither ignored nor collapsed to a constant.

5.3 The representation is mostly not the deciding factor

Q2 is where the answer is uncomfortable, and we report it in full. Across the four families of Table 2, at a matched budget (58,550 against 58,700 specification-encoder parameters, 0.26 % apart) and with both arms’ hyperparameters chosen on the same seen-specification validation split:

- on **reaction** ($\rho \leq 9$) the structured encoder is ahead on every held-out axis by three to four times the threshold — 0.4080 against 0.5754 on the held-out Gaussian family — and the difference is significant under a one-sided rank test on all four ($p \leq 0.045$, Table 4);
- on **convection–reaction** ($\rho \leq 5$) it is better on all six columns, but by less than the threshold on five of them and not significantly on any ($p \geq 0.075$);
- on **reaction–diffusion** the two are indistinguishable, and on the held-out Gaussian family the control is very slightly ahead (0.2585 against 0.2513);
- on **convection–reaction** at $\rho \leq 9$ the control is ahead on every column (0.3356 against 0.4148).

The one apparent win does not survive a sibling control. The control in Table 2 reads the profile at $M = 32$ sensors, the value its own validation sweep selected. Doubling that to $M = 64$ — still a raw sampled profile, still no structure, re-matched to 58,550 parameters at 58,537 (0.02 % apart) — removes the reaction-family advantage entirely. On the 10 seeds all three arms share, the denser control reaches 0.0920 on the seen specifications against 0.1646 for us, 0.3604 against 0.3691 on the held-out Gaussian family, and 0.2870 against 0.3115 on the unseen combinations; it collapses 1/10 times against our 2/10 (and the $M = 32$ control’s 5/10). A paired test favours us over the selected control ($p = 0.019$ on the Gaussian family) and over the denser one not at all ($p = 0.116$).

We therefore withdraw the claim. Across four families and two configurations of the control, **the structured representation does not significantly beat a well-configured sampled-value representation on any axis of any family**; it is nominally ahead on convection–reaction ($p \geq 0.075$), level on reaction–diffusion, and behind on stiff convection–reaction. What looked like a win was a comparison against one particular — validation-selected, budget-matched, but under-sensored — setting of the control. We leave Table 4 unemphasised for that reason, and we would not have found this had the denser control not been run at full seed count.

On our own decision rule. We pre-registered two criteria: a margin of at least $\delta = \max(0.01, 0.1 \times \text{control})$, and a margin exceeding the pooled across-seed standard deviation. Applied literally, the second is never met on any axis of any family — partly because the reaction outcome is *bimodal* (a run converges or collapses), which makes a standard deviation the wrong dispersion statistic. We report the margin, δ , the pooled s.d. and a rank test side by side rather than choosing whichever reading flatters the method, and, given the paragraph above, we claim no representational win at all.

Two mechanism hypotheses, both refuted. Before the denser control had finished, the reaction result looked real and we sought a mechanism for it. Across the coefficient points of the first three families the margin correlated with the reaction coefficient at Spearman +0.74, and within each family at +1.00, +0.94 and +0.88. That predicts a large margin for a convection-bearing family pushed to $\rho = 9$; we ran one, and the margin turned *negative*. Pooled over all 46 coefficient points of all four families the correlation with ρ is 0.014 (Figure 5, Table 5). An earlier hypothesis, that the payoff tracks how much of the initial *shape* survives the horizon, fares no better (−0.001). Both are reported as refuted, and in hindsight both were explanations for an effect that was not there.

The representation ablations locate what does and does not help. Table 14 varies the structured representation at a matched budget. On convection–reaction, replacing amplitude-weighted pooling by an unweighted sum costs 0.3744 against 0.3055 on the held-out Gaussian family, and flattening the identical features into a single MLP — keeping the features, discarding the shared-per-mode processing — costs 0.3587. Those two design choices are therefore doing the work they were meant to do. The PDE-symbol features are not: deleting them changes nothing (0.3032), and on reaction they help (0.4527 without, 0.4080 with) while unweighted pooling is there the *better* choice (0.3068). Meanwhile doubling the control’s sensor count from 32 to 64 — a change that adds no structure whatsoever — moves it by a comparable amount (0.3257 $\rightarrow$ 0.3394). The structured components matter relative to each other and inconsistently across families; they do not separate the structured family from a well-tuned unstructured one.

6 What the encoder represents

Section 5.3 leaves an apparent contradiction: both representations are used, both generalise, and neither is reliably more accurate than the other — yet they are not the same object. This section asks what actually differs between the codes they produce, and ends up explaining why the difference does not reach the error. Three diagnostics are run on the trained models, all on the *same* specification sets used above.

Non-degeneracy. The encoding is used and is not collapsed. Decomposing the latent variance over the trained specifications into the part explained by the initial condition and the part explained by the coefficients shows that the initial condition accounts for 69.2 % of it in the structured encoder (79.1 % in the control), so the profile is genuinely represented rather than ignored. The conditioning-removed evaluation in Table 14 is the behavioural counterpart: replacing the encoded profile by another

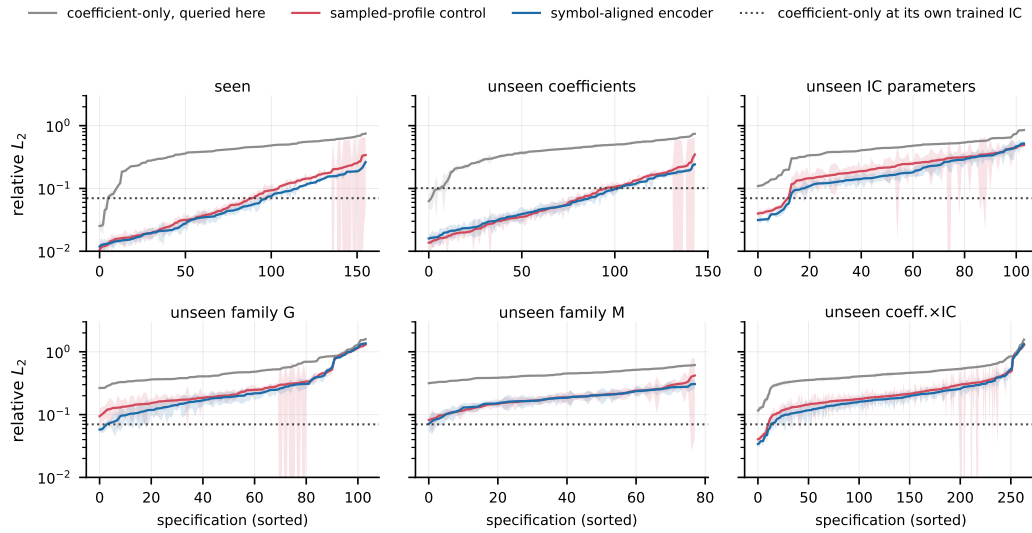


Figure 4: Per-specification error curves, all three arms (specifications sorted by error within each panel; shaded band is the across-seed s.d.). Grey: the coefficient-only surrogate *queried* at each specification — it is not a function of the profile, so it returns its trained solution and sits far above both conditioned arms in every panel, including on the seen specifications. The dotted line is that same model’s accuracy at its own trained initial condition, the best it can reach without retraining. The two conditioned arms are trained identically and differ only in how the specification is represented; they track each other closely, which is the finding of §5.3.

specification’s profile, by a constant, or by the reference training profile — in every case leaving the coefficients correct — degrades held-out accuracy.

6.1 The structured code carries more about an unseen functional family

The clearest representational difference is not in the solution error but in what the latent retains. Fit a ridge probe on the *trained* specifications from $\mathbf{h}_{\text{spec}}$ to a description of the initial condition — its level, scale, spectral centroid and the first six Fourier components of its unit-RMS shape — and score it on held-out profiles against the training prior, so that $R^2 = 0$ means the latent adds nothing to what the training distribution already implies. On the held-out Gaussian family the structured code scores 0.615 against 0.335 for the control on convection–reaction, 0.664 against 0.372 on reaction, and 0.643 against 0.256 on reaction–diffusion: consistently about twice as much decodable information about a functional family it was never trained on. On the held-out *parameters* of the trained family the two are comparable, which is what one expects — there, interpolation suffices. On the three-mode family the ordering reverses (0.017 against 0.246); amplitude-weighted pooling sums the modal contributions, and a superposition of three modes is exactly the case where individual modal components become harder to read back out of the sum.

6.2 Distance in the latent predicts held-out error — but so does distance between profiles

For every held-out specification we measure the distance from its code to the nearest trained specification’s code, normalised by the median pairwise distance among trained specifications, and correlate that with the error the model actually makes there. The correlation is positive for both arms and larger for the structured code on two of three families (0.599 against 0.507; 0.335 against 0.194; 0.393 against 0.351). But the naive baseline for this diagnostic — simply measuring the L_2 distance between the raw profiles — does about as well (0.603, 0.422, 0.413). The latent is organised, and its geometry is predictive, but it is not more predictive than the input geometry it was built from. We report this as a null result for the diagnostic, not as evidence for the encoder.

6.3 The decoder, not the encoder, is the binding constraint

Table 7 is the reason the representation matters so little. It compares the zero-shot conditioned models against the thing the surrogate exists to replace: a coefficient-only P²INN *retrained from scratch*

Table 5: Where the *representation* pays off. For each coefficient point of each core family, the margin of the symbol-aligned encoder over the matched sampled-profile control on the held-out Gaussian family, against the reaction coefficient. Spearman correlations over all 46 points are given below.

family	ρ	ours	control	margin
reaction	1	0.4938	0.4702	-0.0236
	3	0.3404	0.4480	+0.1075
	5	0.4227	0.6052	+0.1826
	7	0.4025	0.6671	+0.2647
	9	0.3808	0.6865	+0.3057
conv.-reac.	1	0.4176	0.4121	-0.0055
	2	0.3403	0.3507	+0.0104
	3	0.2877	0.3041	+0.0163
	4	0.2488	0.2688	+0.0200
	5	0.2257	0.2823	+0.0566
reac.-diff.	1	0.4163	0.3660	-0.0504
	2	0.2888	0.2718	-0.0170
	3	0.2234	0.2298	+0.0063
	4	0.1839	0.1985	+0.0147
	5	0.1654	0.1798	+0.0144

Spearman(margin, ρ) = 0.01 Spearman(margin, ν) = 0.04 Spearman(margin, β) = -0.32
Spearman(margin, shape survival) = -0.00 (the hypothesis this family was added to test; rejected)

Table 6: Latent diagnostics of the specification encoder. R^2 is that of a ridge probe fitted on the *trained* specifications and evaluated on held-out ones; Spearman ρ correlates held-out error with distance to the nearest trained specification, measured in the encoder’s latent and, for comparison, between the raw profiles.

diagnostic	sampled-profile control	symbol-aligned encoder
latent variance explained by the initial condition (%)	79.1	69.2
latent variance explained by the coefficients (%)	9.6	19.1
probe R^2 , held-out IC parameters	0.711	0.600
probe R^2 , held-out family G	0.335	0.615
probe R^2 , held-out family M	0.246	0.017
ρ (error, latent distance)	0.507	0.599
ρ (error, profile distance)	0.532	0.603

on the held-out profile itself, at the same budget. On the reference zoo Gaussian `gauss` ($\sigma = \pi/4$) the retrained oracle reaches 0.6237 and the zero-shot structured model 0.7508; on `gauss_pi_2`, 0.1825 against 0.2768; on an unseen sinusoid, 0.2186 against 0.2254. Zero-shot conditioning costs between 0.007 and about 0.13 relative L_2 compared with retraining — but the retrained model itself is only that accurate. At this budget a 50-wide residual decoder simply cannot represent a localised bump under strong reaction much better than that, no matter what code it is handed. Two further observations point the same way: the errors of the two conditioned arms are strongly correlated across specifications, and the seen-specification validation errors of the two arms after the hyperparameter sweep are 0.1220 and 0.1187 — statistically indistinguishable before a single held-out set is touched.

Reading. The structured code demonstrably *contains* more about an unseen functional family (the probe), and it is demonstrably *used* (the conditioning-removal ablation) — yet it does not produce lower solution error. Both facts are consistent only if the decoder cannot spend the extra information, which is what the oracle says. The one place in this study where the representation changes an outcome is the transfer to a boundary encoding absent from training (§8.1) — and even there the latent diagnostic of Figure 10 shows the two codes are placed equally well, so what differs is what the code *contains*, not how it is organised.

7 Where it stops working

Generalisation has to end somewhere, and the mechanism of §6 predicts where: the shared per-mode network transfers to modes it never saw excited in isolation only insofar as those modes’ tokens

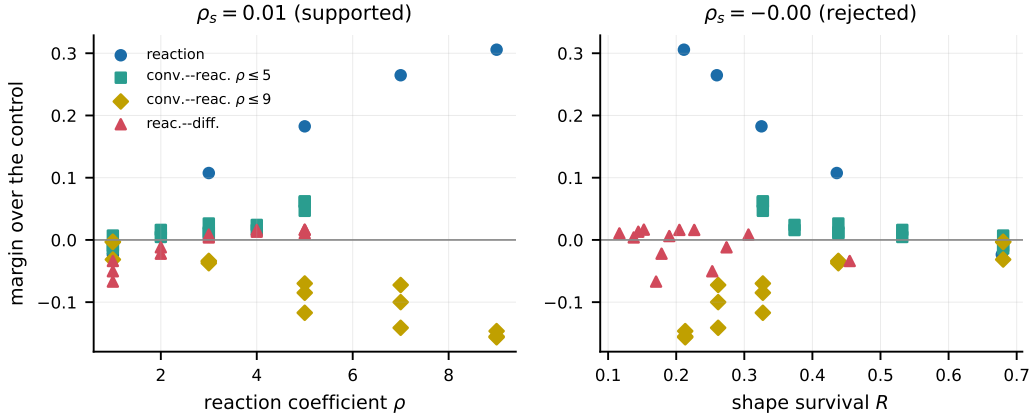


Figure 5: Two mechanism hypotheses, both refuted. Each point is one coefficient setting of one core family; the vertical axis is the margin of the structured encoder over the matched control on the held-out Gaussian family. Left: against the reaction coefficient — the relationship holds within each of the first three families but not across the fourth, run to test it. Right: against shape survival.

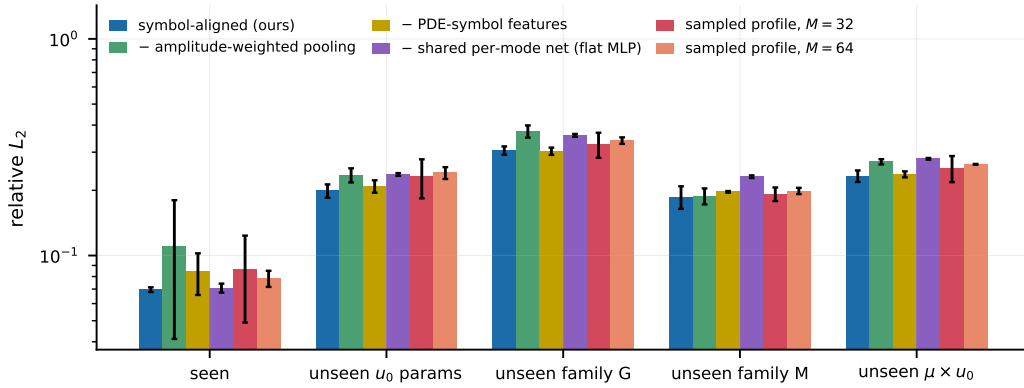


Figure 6: Representation ablations at a matched specification-encoder budget (convection–reaction), together with the conditioning-removed evaluation.

resemble tokens it did see, and the level/scale channels transfer only over the range they were exercised on. We push all three axes well past the trained region on the reaction–diffusion family — a family whose dependence on the initial condition is nonlinear, so a failure here cannot be dismissed as an artefact of a linear solution operator.

The reference is not the failure. Every specification in these sweeps was checked for reference convergence before being used (`logs/ood_reference_validation.json`): refining the spatial grid from 256 to 512 changes the reference by at most 1.3×10^{-15} in relative L_2 over all 20 sweep points, halving the time step by at most 0.0077, and an independent stiff method-of-lines integration of the same semi-discrete system differs from it by at most 0.0110. Any error above that level is the model’s.

7.1 A generalisation boundary in initial-condition sharpness

The sharpness axis produces a clean, monotone and severe failure. Gaussian bumps of width σ inside the trained band cost 0.1188 at $\sigma = 1.5$; narrowing to $\sigma = 0.7$ costs 0.3135, to $\sigma = 0.35$ costs 0.5911, and to $\sigma = 0.25$ costs 0.7241 — a factor of six, with the prediction degenerating into a smoothed version of the true bump. The initial-condition scale axis behaves the same way more gently (0.1679 at scale 1 rising to 0.6096 at scale 16). The wavenumber axis does *not* fail: error rises from 0.1218 at $k = 1$ to 0.2425 at $k = 4$ and then plateaus (0.2576 at $k = 12$), so the shared per-mode network

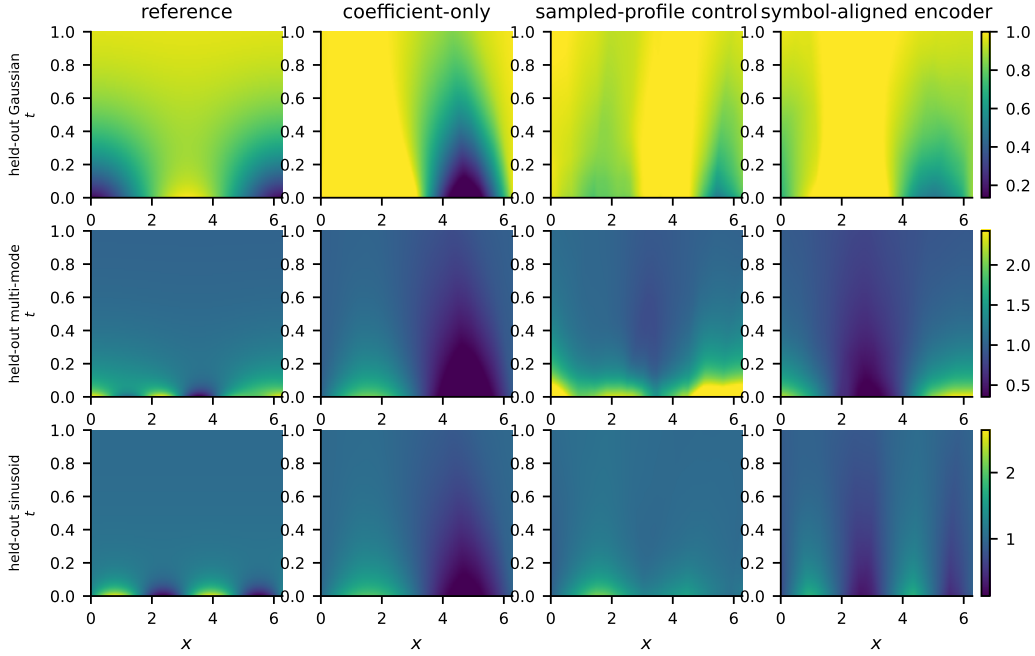


Figure 7: Predicted fields at $(\nu, \rho) = (3, 3)$ for three held-out initial conditions, against the reference. The coefficient-only surrogate returns its trained solution regardless of the query.

Table 7: The price the surrogate exists to avoid. *Retrained oracle*: a coefficient-only P²INN trained from scratch on the held-out profile itself, at the same budget. The conditioned models are queried *zero-shot*. Reaction-diffusion, trained coefficients, mean over 3 seeds (oracle) and 5 seeds (conditioned).

held-out initial condition	retrained oracle	sampled profile	symbol-aligned
gauss ($\sigma=\pi/4$, zoo)	0.6237	0.7028	0.7508
gauss_pi_2 ($\sigma=\pi/2$, zoo)	0.1825	0.2280	0.2768
three-mode (family M)	0.1056	0.2157	0.2120
unseen sinusoid (family S)	0.2186	0.2194	0.2254

does transfer to modes it never saw excited in isolation — it is amplitude concentration in space, not frequency content, that breaks the surrogate.

This is not under-training. We re-ran both conditioned arms with a $3\times$ wider specification-encoder path ($H = 100 \rightarrow 300$; 475,550 parameters against 58,550, with the control matched to 474,815), $3\times$ the epochs (20,000 $\rightarrow$ 60,000) and $2\times$ the collocation points per specification (1000 $\rightarrow$ 2000), holding the decoder, the coordinate encoder, the loss and the seeds fixed. The failure is untouched: at $\sigma = 0.25$ the enlarged structured model reaches 0.6694 against 0.7241 at the reference budget, and the enlarged control 0.6628 against 0.6114. Nine times the training compute and eight times the encoder capacity buy a few per cent.

This is not the reference either. Every specification in these sweeps was checked for reference convergence before it was used (logs/ood_reference_validation.json). Over all 20 sweep points, doubling the spatial grid changes the reference by at most 1.3×10^{-15} relative L_2 , halving the time step by at most 0.0077, and an independent stiff method-of-lines integration of the same semi-discrete system differs by at most 0.0110 — two orders of magnitude below the errors above.

But it is not specific to our encoder. The matched control fails on the same axis, at the same place, by nearly the same amount. Under the discipline we set ourselves this is a boundary of the *parametric-PINN surrogate at this budget* on a family whose dependence on the initial condition is

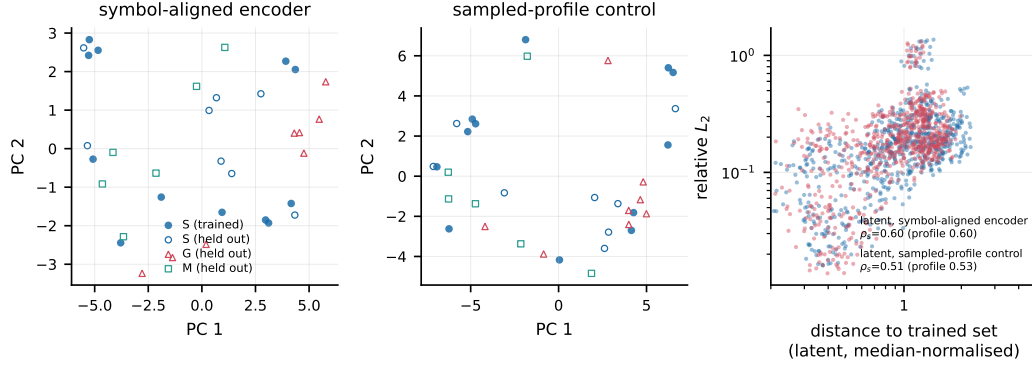


Figure 8: Left, centre: the first two principal components of $\mathbf{h}_{\text{spec}}$ at a fixed coefficient pair, for every trained (filled) and held-out (open) initial condition. Right: held-out error against the distance from the query specification to the nearest trained specification, measured in each model’s own latent (normalised by the median pairwise distance among trained specifications).

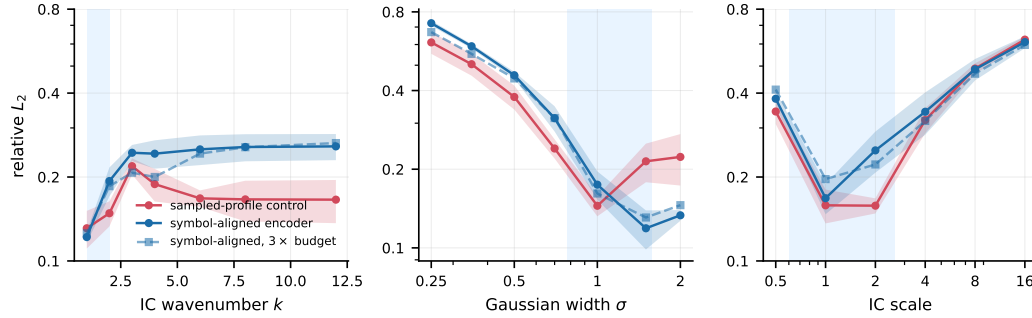


Figure 9: Out-of-distribution sweeps along three axes of the initial-condition space, evaluated on the *same* trained models (shaded band: the trained region; error bars: across-seed s.d.). Left: sinusoids of increasing wavenumber. Centre: Gaussian bumps of decreasing width. Right: profiles of increasing scale. The dashed curve is the symbol-aligned encoder retrained under a $3\times$ encoder width, $3\times$ epochs and $2\times$ collocation budget.

genuinely nonlinear — not a boundary of the structured encoding in particular, and we do not claim it as one. It is consistent with §6: a narrow bump asks the 50-wide decoder for a spatial feature it cannot render, and no encoding of the input changes that.

8 Breadth: boundary conditions and PDE families

Table 8 summarises what a single conditioned model is asked to span across the whole study. The two subsections below report the two axes that the coefficient-only surrogate cannot represent at all.

8.1 One model, three boundary specifications

The boundary condition enters the encoder twice: as a type embedding in the global token, and — more importantly — by selecting the eigenbasis in which the profile is expanded (Fourier / sine / cosine). It enters the objective through the boundary term, which is the periodic mismatch, the Dirichlet level residual or the Neumann derivative residual as appropriate. Reference solutions for all three boundary types come from the authored split-step solver of §4.2, validated to 1.1×10^{-4} relative L_2 against an independent stiff method-of-lines integration and to 4.2×10^{-15} in max-norm against the shipped spectral reference on the periodic overlap.

An important caveat, measured. Homogeneous Neumann data requires $u'_0(0) = u'_0(2\pi) = 0$, so the boundary study restricts the profiles to even-symmetric ones. An even-symmetric 2π -periodic profile, however, *already* satisfies the Neumann conditions — so on this profile family the periodic

Table 8: Breadth: what a single conditioned model is asked to span. Every held-out set is evaluated and never trained on, and no held-out set was used to select any hyperparameter.

quantity	count	detail
initial-condition <i>functional</i> families spanned by one model	3	1 trained (S: $c + A \sin(kx + \varphi)$), 2 held out entirely (G: Gaussian bumps, M: three-mode)
distinct initial profiles evaluated	34	12 trained, 8 held-out parameters, 8 family G (incl. the 2 reference-zoo Gaussians), 6 family M
boundary specifications spanned by one model	3	periodic, Dirichlet, Neumann — two distinct boundary-value problems (§8.1)
PDE families spanned by one model	5	convection, diffusion, reaction, reaction–diffusion, CDR
PDE families used as separate controlled comparisons	4	reaction, reaction–diffusion, convection–reaction at $\rho \leq 5$ and at $\rho \leq 9$
held-out axes evaluated per family	5	unseen coefficient; unseen IC parameters; two unseen IC families; unseen coefficient $\times$ profile
specifications in the largest single model	552	552 (five PDE families) and 468 (three boundary specifications), against 13 for the coefficient-only baseline
specifications each model is <i>evaluated</i> at	1668	up to 1668; every held-out set is evaluated, never trained on

Table 9: One model conditioned on the boundary condition (reaction–diffusion; references from the authored, validated non-periodic solver). P/D/N is trained on all three boundary specifications; P/D is trained on periodic and Dirichlet data only, so for it every Neumann column is a boundary encoding absent from training in every form. Note that on the even-symmetric profiles that Neumann compatibility requires, the Neumann and periodic solutions coincide (to 3×10^{-5}); the Neumann columns therefore test the boundary *encoding*, not new physics.

boundary	method	seen	uns. IC par.	uns. fam. G	uns. comb.
periodic	sampled profile (P/D/N)	0.1049 $\pm$ 0.0074	0.1407 $\pm$ 0.0234	0.2691 $\pm$ 0.0411	0.2022 $\pm$ 0.0314
	symbol-aligned (P/D/N)	0.0968 $\pm$ 0.0076	0.1287 $\pm$ 0.0091	0.2694 $\pm$ 0.0093	0.1961 $\pm$ 0.0069
	sampled profile (P/D)	0.1845 $\pm$ 0.0371	0.2000 $\pm$ 0.0333	0.3090 $\pm$ 0.0027	0.2524 $\pm$ 0.0164
	symbol-aligned (P/D)	0.1096 $\pm$ 0.0013	0.1201 $\pm$ 0.0046	0.3030 $\pm$ 0.0092	0.2087 $\pm$ 0.0040
dirichlet	sampled profile (P/D/N)	0.0918 $\pm$ 0.0112	0.1385 $\pm$ 0.0226	0.4183 $\pm$ 0.0642	0.2753 $\pm$ 0.0410
	symbol-aligned (P/D/N)	0.0889 $\pm$ 0.0089	0.1317 $\pm$ 0.0103	0.5317 $\pm$ 0.0179	0.3287 $\pm$ 0.0047
	sampled profile (P/D)	0.1034 $\pm$ 0.0116	0.1435 $\pm$ 0.0116	0.4605 $\pm$ 0.0077	0.2996 $\pm$ 0.0092
	symbol-aligned (P/D)	0.0962 $\pm$ 0.0151	0.1296 $\pm$ 0.0135	0.5122 $\pm$ 0.0428	0.3185 $\pm$ 0.0136
neumann	sampled profile (P/D/N)	0.1043 $\pm$ 0.0074	0.1402 $\pm$ 0.0229	0.2691 $\pm$ 0.0393	0.2018 $\pm$ 0.0303
	symbol-aligned (P/D/N)	0.0960 $\pm$ 0.0072	0.1282 $\pm$ 0.0100	0.2662 $\pm$ 0.0073	0.1943 $\pm$ 0.0068
	sampled profile (P/D)	0.2751 $\pm$ 0.0199	0.2764 $\pm$ 0.0154	0.3117 $\pm$ 0.0045	0.2931 $\pm$ 0.0074
	symbol-aligned (P/D)	0.2158 $\pm$ 0.0238	0.1978 $\pm$ 0.0325	0.3203 $\pm$ 0.1004	0.2589 $\pm$ 0.0550

and Neumann boundary-value problems have the same solution. We measured it: the two reference fields agree to 3×10^{-5} in max-norm on a common grid, while periodic and Dirichlet differ by 0.99. The study therefore covers **three boundary specifications** — three grids, three transforms inside the encoder, three boundary loss terms — but **two distinct boundary-value problems**. We state the result at that strength.

A single model trained on all three specifications reaches 0.0968, 0.0889 and 0.0960 relative L_2 on the seen specifications under periodic, Dirichlet and Neumann data — so conditioning on the boundary costs essentially nothing relative to a periodic-only model at the same budget (0.1225 for the periodic-only reaction–diffusion model of Table 2). It generalises within each boundary specification to profiles held out of training: on the held-out Gaussian family, 0.2694 (periodic), 0.2662 (Neumann) and 0.5317 (Dirichlet), and to held-out coefficient $\times$ profile compositions under each (Table 9). Dirichlet is the hard case: pinning the ends at $u_0(0)$ makes the solution depend on a feature of the profile — its boundary value — that the Gaussian family varies over a range the trained cosines never visited.

An unseen boundary encoding. The P/D rows are trained on periodic and Dirichlet data only and then queried under Neumann data: an unseen grid, an unseen transform, an unseen boundary term and a one-hot component that was never active. Because the Neumann field *is* the periodic field

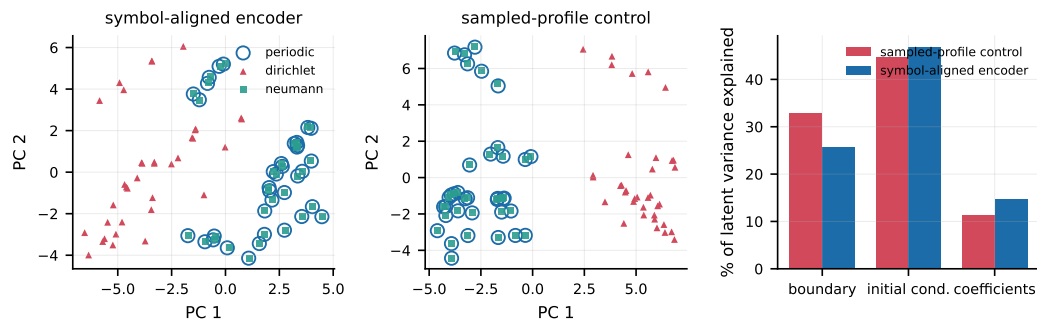


Figure 10: How the boundary specification is represented. Left, centre: the first two principal components of h_{spec} for the model trained on all three boundary specifications, at six coefficient pairs and six profiles each. Periodic is drawn as a large open ring so that its coincidence with Neumann is visible rather than hidden by overplotting. *Both* encoders place every Neumann code exactly on top of its periodic counterpart — which is correct, since on this profile family the two are the same boundary-value problem — and separate Dirichlet cleanly. Right: the share of latent variance attributable to each part of the specification.

here, the natural yardstick is what the same model scores on the periodic problem, 0.1096. Both arms fall short of it, so both are disturbed by the unfamiliar encoding — but the structured encoder much less: 0.2158 against 0.2751, a 22 % reduction — and, now that the reaction-family margin has been withdrawn (§5.3), the only place in this study where the structured representation changes an outcome. Whether that is the mechanism its design predicts is a question the latent can answer, and the answer is partly no.

What the latent shows, and what it does not. Both encoders devote a large share of their latent to the boundary — 25.7 % of the variance for ours and 32.9 % for the control, against 46.9 % and 44.8 % for the initial condition — so the boundary is genuinely represented in both. More tellingly, fit a linear map from the latent to a boundary indicator using *only* periodic and Dirichlet codes and ask where it places a Neumann code: ours scores it 0.998 periodic and 0.002 Dirichlet, and the control scores it 0.998 and 0.002 — identically. Both correctly recognise the Neumann problem as the periodic one in disguise, and Figure 10 shows the same thing geometrically.

The advantage is therefore *not* explained by better placement of the boundary in the latent: on that diagnostic the two arms are indistinguishable. What remains is the *content* rather than the position of the code. Under Neumann data our encoder recomputes the modal coefficients in the cosine basis on the Neumann grid, which is the correct expansion of that field whether or not the boundary type was ever trained on; the control reads raw profile values at sensor locations on a half-cell-shifted grid it has never seen, and its only signal that anything has changed is a one-hot component that was inert throughout training. We report the advantage as measured (22 % error reduction) and the mechanism as partly open, rather than asserting a basis-selection story that the latent diagnostic does not support.

8.2 One model, five PDE families

Trained jointly on convection, diffusion, reaction, reaction–diffusion and convection–diffusion–reaction — 5 families spanning 46 coefficient points and 12 profiles in one network — the model reaches 0.2895, 0.1698, 0.0887, 0.0947 and 0.0620 relative L_2 on the seen specifications of each, and generalises within each family to the held-out initial-condition families and to held-out coefficient×profile compositions (Table 10). The two representations are again close.

An honest negative: transfer to an unseen PDE type. The NO- β rows are trained on diffusion, reaction and reaction–diffusion only, so $\beta = 0$ throughout training and convection and CDR are PDE *types* the model has never met. Queried on convection it reaches 0.5762 (against 0.2895 when convection is in the training mixture) and on CDR 0.3266 (against 0.0620) — a factor of 2.0–5.3× worse, i.e. no useful transfer. The control fails identically (0.5957, 0.3292).

The mechanism is visible in the encoder. The convective entries of the per-mode token are $\cos(\beta q_k T)$, $\sin(\beta q_k T)$ and a bounded $\beta q_k T$; with $\beta \equiv 0$ these are constant across every training specification, their standardised value is fixed, and the weights attached to them are never excited. At test time

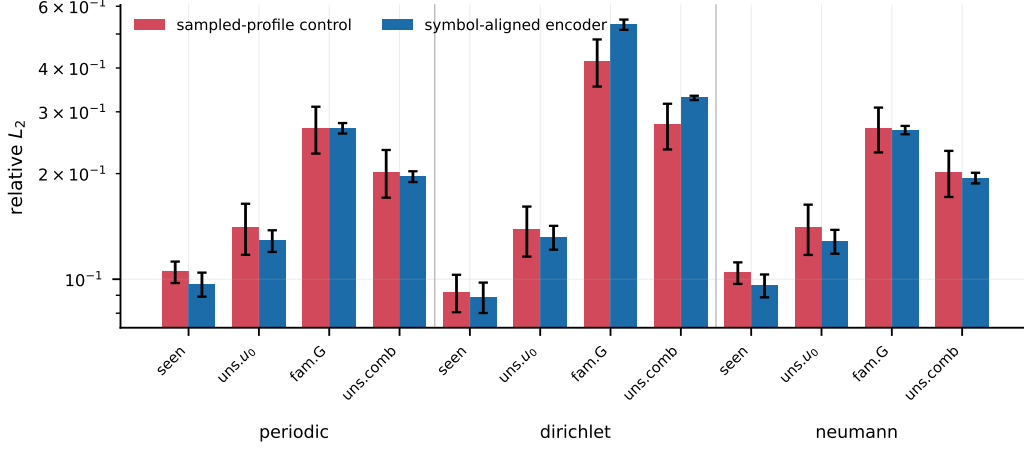


Figure 11: One model conditioned on the boundary specification (reaction–diffusion). Periodic and Neumann coincide as boundary-value problems on the even-symmetric profiles Neumann compatibility requires; they differ as encodings.

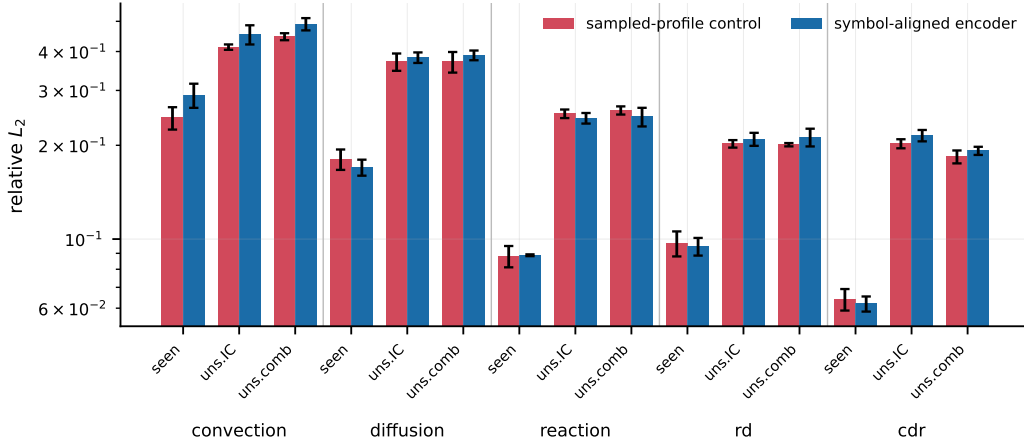


Figure 12: A single model trained jointly on five PDE families, evaluated per family on the seen specifications, on held-out initial-condition families, and on held-out coefficient $\times$ profile compositions.

$\beta \neq 0$ moves an input direction along which the network has no trained response. We tried a representational fix: re-expressing the modal phase in the frame co-moving with the convective rotation, $\varphi_k \rightarrow \varphi_k - \beta q_k T$, so that a non-zero β *reparameterises* an input the encoder has seen at every value (phase) rather than exciting a constant one. It reaches 0.5794 on convection — no better. We conclude the obstruction is not in the encoder at all: the *solution* translates in time, the decoder and its coordinate encoder have never produced a travelling field, and no re-encoding of the input can supply that. Both attempts are documented in `attempts_log.md`.

9 Related work

Parametric PINN surrogates. PINNs [Raissi et al., 2019] solve one specification per network and fail outright in convection- and reaction-dominated regimes [Krishnapriyan et al., 2021]. Amortising over the coefficient vector is the standard remedy. P²INN [Cho et al., 2024] — whose factorisation, decoder, loss and budget we adopt unchanged — encodes μ explicitly rather than appending it to the coordinates, and shows that the explicit encoding, not the extra input, is what buys the accuracy; the same conclusion is what we are after one level up. Hypernetwork variants generate PINN weights from the coefficients [Ha et al., 2017, de Avila Belbute-Peres et al., 2021], and meta-learning variants produce good initialisations for fast per-instance adaptation [Huang et al., 2022, Penwarden et al.,

Table 10: A single model over five PDE families (ALL), and the same setup trained on diffusion, reaction and reaction–diffusion only (NO- β), for which convection and CDR are PDE types absent from training in every form. “uns. IC” is the held-out Gaussian and multi-mode families; “uns. comb.” is a held-out coefficient with a held-out profile.

family	method	seen	uns. IC	uns. comb.
convection	sampled profile (ALL)	0.2448 $\pm$ 0.0202	0.4133 $\pm$ 0.0082	0.4464 $\pm$ 0.0115
	symbol-aligned (ALL)	0.2895 $\pm$ 0.0258	0.4535 $\pm$ 0.0321	0.4898 $\pm$ 0.0222
	sampled profile (NO- β)	0.5957 $\pm$ 0.0815	0.6420 $\pm$ 0.0774	0.6735 $\pm$ 0.0671
	symbol-aligned (NO- β)	0.5762 $\pm$ 0.0272	0.5831 $\pm$ 0.0862	0.6349 $\pm$ 0.0841
diffusion	sampled profile (ALL)	0.1803 $\pm$ 0.0136	0.3705 $\pm$ 0.0237	0.3704 $\pm$ 0.0282
	symbol-aligned (ALL)	0.1698 $\pm$ 0.0100	0.3825 $\pm$ 0.0149	0.3890 $\pm$ 0.0140
	sampled profile (NO- β)	0.2770 $\pm$ 0.0069	0.4421 $\pm$ 0.0462	0.4389 $\pm$ 0.0549
	symbol-aligned (NO- β)	0.2946 $\pm$ 0.0050	0.4652 $\pm$ 0.0587	0.4627 $\pm$ 0.0601
reaction	sampled profile (ALL)	0.0881 $\pm$ 0.0069	0.2525 $\pm$ 0.0080	0.2588 $\pm$ 0.0078
	symbol-aligned (ALL)	0.0887 $\pm$ 0.0006	0.2444 $\pm$ 0.0096	0.2469 $\pm$ 0.0170
	sampled profile (NO- β)	0.0563 $\pm$ 0.0047	0.2558 $\pm$ 0.0164	0.2572 $\pm$ 0.0108
	symbol-aligned (NO- β)	0.0550 $\pm$ 0.0028	0.2509 $\pm$ 0.0020	0.2478 $\pm$ 0.0193
reac.–diff.	sampled profile (ALL)	0.0969 $\pm$ 0.0089	0.2022 $\pm$ 0.0056	0.2009 $\pm$ 0.0025
	symbol-aligned (ALL)	0.0947 $\pm$ 0.0062	0.2092 $\pm$ 0.0101	0.2122 $\pm$ 0.0139
	sampled profile (NO- β)	0.1330 $\pm$ 0.0041	0.2202 $\pm$ 0.0204	0.2157 $\pm$ 0.0172
	symbol-aligned (NO- β)	0.1431 $\pm$ 0.0097	0.2146 $\pm$ 0.0087	0.2148 $\pm$ 0.0153
conv.–diff.–reac.	sampled profile (ALL)	0.0640 $\pm$ 0.0051	0.2024 $\pm$ 0.0068	0.1837 $\pm$ 0.0088
	symbol-aligned (ALL)	0.0620 $\pm$ 0.0035	0.2149 $\pm$ 0.0090	0.1920 $\pm$ 0.0058
	sampled profile (NO- β)	0.3292 $\pm$ 0.1659	0.3870 $\pm$ 0.1428	0.3837 $\pm$ 0.1810
	symbol-aligned (NO- β)	0.3266 $\pm$ 0.1347	0.3508 $\pm$ 0.0936	0.3479 $\pm$ 0.1224

2023]; recent pseudo-time parametric PINNs [Wang et al., 2025] likewise condition on coefficients only. In every case the initial and boundary conditions live in the loss, so a query at a new profile is a retraining job. That is the gap we close.

Operator learning. DeepONets [Chen and Chen, 1995, Lu et al., 2021] and neural operators [Li et al., 2021, Kovachki et al., 2023, Bhattacharya et al., 2021] learn maps between function spaces and are conditioned on the input function by construction — the branch network reads it as a vector of sensor values. Physics-informed DeepONets [Wang et al., 2021] train that map from the residual, exactly the regime we are in. Our matched control *is* that design, transplanted into the parametric-PINN factorisation, and the experiments below say that in the few-training-profile regime the representation, not the conditioning per se, is what decides whether the model generalises off the trained profile manifold. Spectral variants of the branch network exist — Fourier and Chebyshev embeddings of the input function are used to combat spectral bias — but they embed the function alone; the token in Eq. (5) instead carries the *joint* coefficient–mode object, the symbol of the operator acting on that mode, which is what makes unseen coefficient $\times$ profile combinations behave. Transformer neural operators for initial–boundary value problems [Bora et al., 2024] and multi-physics pretraining [McCabe et al., 2023, Herde et al., 2024, Takamoto et al., 2022] scale this up with data; we work in the data-free, physics-informed, single-GPU regime, and our question is representational rather than about scale.

Conditional neural fields. Treating a solution as a coordinate network modulated by a latent code is the neural-field view [Park et al., 2019, Sitzmann et al., 2020, Dupont et al., 2022, Perez et al., 2018], imported into PDE surrogates by CORAL [Serrano et al., 2023] and DINO [Yin et al., 2023], which encode a state by auto-decoding a latent per instance. Those latents are fitted per instance at test time; ours is produced by a feed-forward encoder in one shot and is constrained by an explicit physics-aligned parameterisation. The permutation-invariant pooling in Eq. (6) is a Deep Set [Zaheer et al., 2017] over *modes* rather than over points, and the amplitude weighting is what makes it behave like a superposition rather than a bag.

Boundary conditions as an input. Most PINN work on boundary conditions concerns *enforcement* — hard constraints, adaptive weighting [McClenny and Braga-Neto, 2023], causality-respecting schedules [Wang et al., 2024] — for one fixed boundary type. Conditioning a single model on the

boundary *type* is rarer; the closest is HyPINO [Zappala et al., 2025], which maps a specification — coefficients, sources and Dirichlet/Neumann location and value grids — to PINN weights through a Swin-Transformer hypernetwork, for *linear* 2D elliptic, hyperbolic and parabolic problems. Our treatment differs in kind: we keep the parametric-PINN concatenation factorisation and its budget, we target reaction-bearing *nonlinear* families where the initial-condition dependence is not a linear operator, and the boundary condition is not an extra input grid but the choice of *eigenbasis* in which the profile is expanded, which is what lets one shared per-mode network serve every boundary specification. We also differ in the question asked: our experiments are built to attribute generalisation to structure against a matched control, and to map where it stops.

Positioning. We are not aware of prior work that (i) turns a parametric-PINN coefficient encoder into a joint coefficient/initial/boundary encoder at a matched budget and measures what that buys, or (ii) runs the controlled experiment that separates the *conditioning* from its *representation* — a physics-aligned encoding against a budget-matched sampled-value control, both tuned on the same seen-specification split, across four PDE families, with a conditioning-removal ablation, latent probes, a retraining oracle, and a mechanism hypothesis tested on a family run specifically to falsify it. The individual ingredients — modal expansions, Deep Sets, symbol-aware features, physics-informed operator learning — are all standard. What we contribute is the controlled comparison and its outcome, which is that the conditioning matters a great deal and, at this budget, the representation mostly does not; the literature tends to report the two together and attribute the result to the second.

10 Limitations and conclusions

We set out to extend what a parametric physics-informed surrogate can be conditioned on, and to test whether the way the extra conditioning is represented is what decides its success. The extension itself is unambiguously worth making: one network, at the reference budget, becomes a function of the coefficients, the initial profile and the boundary condition, and answers queries at profiles from functional families it never saw, at boundary types it was not trained on, and across five PDE families — capabilities the coefficient-only surrogate does not have at any accuracy. The evidence that the conditioning is real rather than decorative is behavioural (removing it collapses the accuracy) and representational (most of the latent’s variance tracks the profile, and a probe recovers roughly twice as much about an unseen functional family from the structured code as from the sampled one).

The design lesson is the negative one, and it arrived in two stages. A carefully physics-aligned encoding of the initial condition — one whose every component we can motivate from the operator — does not systematically beat appending sampled values, once the two are given the same capacity, the same budget and the same tuning. Our first reading was softer than that: the encoding appeared to win on the reaction family, and we spent two hypotheses trying to explain where and why, one of which we tested on a purpose-built fifth experiment that refuted it. The stronger and simpler explanation only appeared when the denser-sensor configuration of the *same* control finished at full seed count and matched or beat us on every axis of that family. There was nothing to explain. What we had measured was the difference between our encoder and one particular setting of the control, not between structure and no structure.

The retraining oracle says why there was so little to win: on these benchmarks a 50-wide residual decoder at 20,000 epochs is the binding constraint, and a better description of the input cannot be spent — which is consistent with the probe result, where the structured code demonstrably carries about twice as much about an unseen functional family and it makes no difference to the error. We would expect the ranking between encodings to become informative only once the decoder is no longer saturated. That is a concrete prediction, and one we could not test within the held-fixed budget that makes this a controlled comparison at all.

Limitations. (i) The conditioned models are less accurate than a narrow model at the narrow model’s own specification; we quantify but do not remove this. (ii) The withdrawn reaction-family result illustrates how fragile a representational claim can be: it was significant against the validation-selected control at $p \leq 0.045$ and vanished against a sibling configuration of the same control. Any single-control comparison in this setting should be treated with the same suspicion, including ours. (iii) The generalisation boundary in initial-condition sharpness is a boundary of the surrogate at this budget, shared by the control; we do not claim it as a boundary of the structured encoding. (iv)

Transfer to a PDE type absent from training fails, and a representational fix did not help — we attribute this to the decoder never having produced a travelling field, which the encoder cannot supply. (v) Everything is 1D and the boundary study is confined to the reaction–diffusion family, where the non-periodic reference is well posed for every profile we use.

Outlook. The result we would most like to see tested is the prediction above: run the same controlled comparison with a decoder large enough that the specification code is the limiting factor, and see whether the structured encoding separates from the sampled one the way the probe diagnostics say it should. A second is methodological: a matched-capacity control is only as strong as its own hyperparameters, and a comparison against a single setting of it can manufacture an effect that a second setting removes. Until then the practical advice this study supports is simple — condition the encoder on the whole specification, and do not expect the representation to be where the accuracy comes from.

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Table 11: Validation record for the reference solutions (`logs/reference_validation.json`). Worst case over all tested initial conditions, coefficients and boundary types. V0–V1 establish that the shipped periodic construction is reproduced exactly; V2–V3 that the authored non-periodic solver is converged and agrees with an independent integrator; V4 that the shipped operator-split reference is converged at the sharpest profiles used anywhere here.

check	what is compared	worst case
V0	bridge(zoo profile) vs. shipped solver, all families	0.0 (bit-exact)
V1	authored solver (periodic, shipped Δt) vs. shipped	4.22×10^{-15} (max)
V1b	consistent vs. shipped time-grid convention	0.0022 (rel.)
V2	time refinement $n_{\text{sub}}=1$ vs. 64	0.0189 (rel.)
V2	time refinement $n_{\text{sub}}=4$ vs. 64	0.0050 (rel.)
V2	space refinement $n_x=256$ vs. 512	1.19×10^{-4} (rel.)
V3	split step $n_{\text{sub}}=1$ vs. independent MOL	6.94×10^{-3} (rel.)
V3	split step $n_{\text{sub}}=64$ vs. independent MOL	1.09×10^{-4} (rel.)
V4	shipped reference, $n_x=256$ vs. 512, sharpest held-out ICs	1.53×10^{-7} (rel.)
V4	shipped reference, $n_t=100$ vs. 200, sharpest held-out ICs	0.0047 (rel.)

A Reference solutions and their validation

Periodic reference solutions are produced by the shipped construction executed verbatim; our bridge substitutes only the initial-condition callable and is bit-exact against the unmodified call path (check V0). The authored non-periodic solver keeps the shipped Lie–Trotter scheme — exact pointwise logistic reaction step, exact-in-time diffusion step — and replaces the Fourier eigenbasis of the Laplacian by the sine (Dirichlet) or cosine (Neumann) eigenbasis on the matching grid; under Dirichlet data the constant level $b = u_0(0)$ is subtracted before the transform and restored after, so the diffused deviation satisfies homogeneous data. Checks V1–V3 establish, in order: that in the shipped time convention the authored solver reproduces the shipped spectral reference to machine precision on the periodic overlap; that it converges under time and space refinement; and that it agrees with a completely independent stiff method-of-lines integration of the same semi-discrete system for all three boundary types. Production references use $n_{\text{sub}} = 64$, i.e. the last row of V3. Check V4 is the convergence spot-check the operator-split families require at the sharpest and highest-frequency held-out profiles used anywhere in this study.

B Hyperparameter selection

C Non-regression on the original coefficient-only setting

D Complete per-axis results

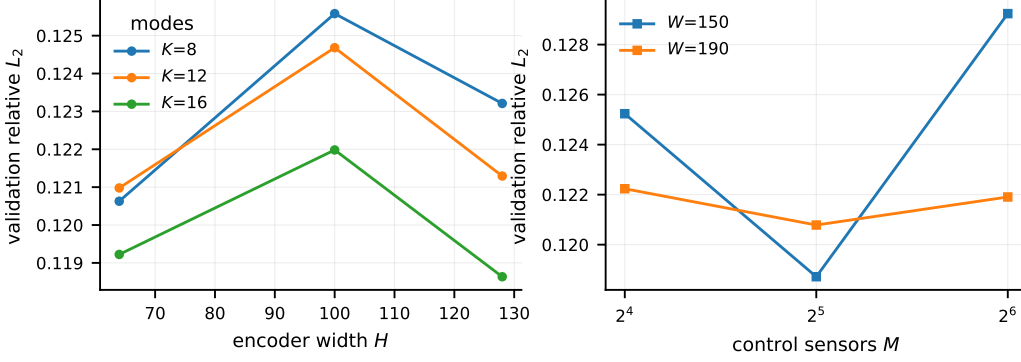


Figure 13: Selection of the conditioning hyperparameters on the *seen-specification* validation split (30 runs, two seeds each; the validation points are the reference sampler’s `idx_val` draw, disjoint from the collocation and test points). No held-out set is involved — the sweep run records contain `val_seen` only. The unconstrained optima are $K = 16$, $H = 128$ (validation 0.1186) and $M = 32$, $W = 150$ (0.1187), but those two configurations differ by 58 % in specification-encoder parameters and so violate the pre-registered $\leq 1\%$ budget-match rule. Applying that rule — keep only pairs whose parameter counts agree to 1 %, then take the best validation error per arm — leaves exactly one feasible pair on the swept grids, and it is what every reported comparison uses: $K = 16$, $H = 100$ (58,550 parameters, validation 0.1220) against $M = 32$, $W = 150$ (58,700 parameters, 0.1187). The constraint costs *us* validation accuracy, not the control.

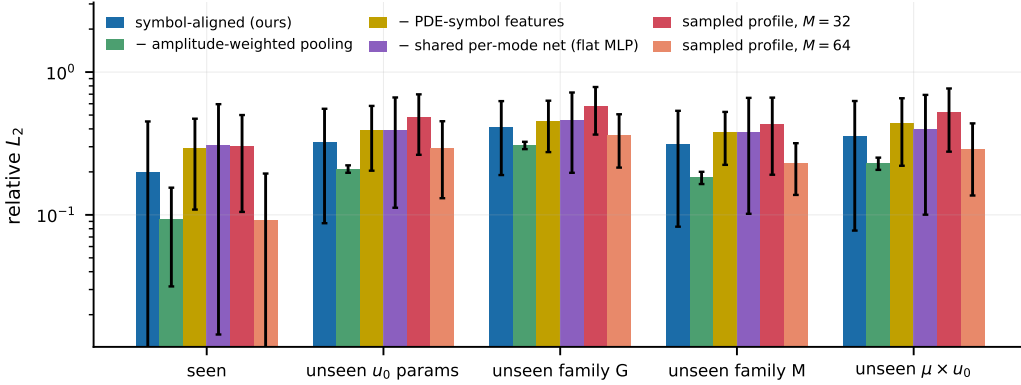


Figure 14: Representation ablations on the reaction family. The denser-sensor control (rightmost bar of each group) is the arm whose full-seed result withdrew the reaction-family claim (§5.3).

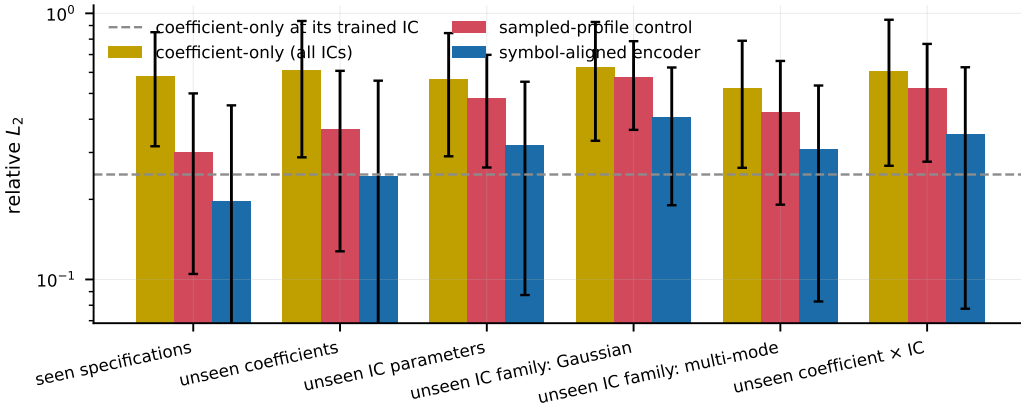


Figure 15: Reaction family, mean relative L_2 per evaluation axis. This is the family on which the structured encoder appeared to win against the validation-selected control; §5.3 shows the advantage does not survive the denser-sensor configuration of that same control.

Table 12: The cost of the extra conditioning on the *original* coefficient-only setting: relative L_2 at the reference initial condition $1+\sin(x)$ under the periodic boundary, per trained coefficient. The baseline is trained on that one profile alone; the conditioned models cover 156 specifications at the same budget. The pre-registered tolerance was $\max(10\%, 0.005)$ over the baseline.

family	(β, ν, ρ)	baseline	control	ours	ours—base
conv.-reac. $\rho \leq 5$	(1,0,1)	0.0264	0.1439	0.1666	+0.1403
	(2,0,4)	0.0816	0.1496	0.1562	+0.0745
	(4,0,2)	0.0468	0.1525	0.1651	+0.1183
	(5,0,5)	0.1125	0.3400	0.1914	+0.0789
	mean	0.0699	0.2017	0.1726	+0.1027
reac.-diff. $\rho \leq 5$	(0,1,1)	0.0734	0.0847	0.0775	+0.0041
	(0,2,4)	0.0761	0.0944	0.0752	-0.0009
	(0,4,2)	0.1329	0.2061	0.2101	+0.0772
	(0,5,5)	0.1079	0.1378	0.1173	+0.0094
	mean	0.1005	0.1419	0.1321	+0.0317
reaction $\rho \leq 9$	(0,0,1)	0.0523	0.0868	0.1176	+0.0652
	(0,0,3)	0.0784	0.1352	0.1764	+0.0979
	(0,0,5)	0.3698	0.5416	0.3934	+0.0235
	(0,0,7)	0.3717	0.7123	0.3903	+0.0186
	(0,0,9)	0.3676	0.7659	0.3809	+0.0133
	mean	0.2480	0.4483	0.2917	+0.0437
conv.-reac. $\rho \leq 9$	(1,0,1)	0.0403	0.2001	0.2336	+0.1933
	(1,0,9)	0.6343	0.3174	0.4836	-0.1507
	(3,0,7)	0.6418	0.3380	0.5005	-0.1413
	(5,0,5)	0.4672	0.2098	0.3561	-0.1110
	mean	0.3997	0.2792	0.3593	-0.0405

Table 13: Representation ablations at a matched specification-encoder budget, and the conditioning-removed evaluation. Every arm is trained with the identical decoder, loss, budget and seeds.

variant	seen	unseen IC par.	unseen fam. G	unseen fam. M	unseen comb.
symbol-aligned (ours)	0.1967 $\pm$ 0.2543	0.3204 $\pm$ 0.2330	0.4080 $\pm$ 0.2180	0.3091 $\pm$ 0.2264	0.3524 $\pm$ 0.2748
— unweighted pooling	0.0933 $\pm$ 0.0618	0.2097 $\pm$ 0.0126	0.3068 $\pm$ 0.0181	0.1823 $\pm$ 0.0179	0.2292 $\pm$ 0.0225
— no symbol features	0.2902 $\pm$ 0.1813	0.3917 $\pm$ 0.1878	0.4527 $\pm$ 0.1776	0.3750 $\pm$ 0.1510	0.4382 $\pm$ 0.2174
— flat MLP on same features	0.3052 $\pm$ 0.2907	0.3881 $\pm$ 0.2759	0.4582 $\pm$ 0.2611	0.3808 $\pm$ 0.2790	0.3962 $\pm$ 0.2959
sampled profile, M=32	0.3025 $\pm$ 0.1977	0.4809 $\pm$ 0.2174	0.5754 $\pm$ 0.2105	0.4267 $\pm$ 0.2357	0.5223 $\pm$ 0.2452
sampled profile, M=64	0.0920 $\pm$ 0.1027	0.2917 $\pm$ 0.1609	0.3604 $\pm$ 0.1461	0.2278 $\pm$ 0.0899	0.2870 $\pm$ 0.1503
<i>conditioning removed at evaluation (symbol-aligned encoder, coefficients kept correct)</i>					
true profile	0.0491 $\pm$ 0.0097	0.1833 $\pm$ 0.0071	0.2695 $\pm$ 0.0054	0.1640 $\pm$ 0.0162	0.1868 $\pm$ 0.0044
shuffled profile	0.3261 $\pm$ 0.0163	0.3261 $\pm$ 0.0166	0.3511 $\pm$ 0.0298	0.2030 $\pm$ 0.0300	0.2782 $\pm$ 0.0053
constant profile	0.3646 $\pm$ 0.0021	0.3152 $\pm$ 0.0205	0.4490 $\pm$ 0.0166	0.1800 $\pm$ 0.0121	0.2568 $\pm$ 0.0173
frozen to $1+\sin(x)$	0.3323 $\pm$ 0.0142	0.3459 $\pm$ 0.0173	0.4257 $\pm$ 0.0219	0.3546 $\pm$ 0.0223	0.3366 $\pm$ 0.0219

Table 14: Representation ablations at a matched specification-encoder budget, and the conditioning-removed evaluation. Every arm is trained with the identical decoder, loss, budget and seeds.

variant	seen	unseen IC par.	unseen fam. G	unseen fam. M	unseen comb.
symbol-aligned (ours)	0.1225 $\pm$ 0.0057	0.1800 $\pm$ 0.0114	0.2585 $\pm$ 0.0121	0.1728 $\pm$ 0.0101	0.2039 $\pm$ 0.0061
— unweighted pooling	0.1206 $\pm$ 0.0093	0.1986 $\pm$ 0.0209	0.2844 $\pm$ 0.0212	0.1665 $\pm$ 0.0112	0.2191 $\pm$ 0.0086
— no symbol features	0.1210 $\pm$ 0.0017	0.2178 $\pm$ 0.0166	0.2635 $\pm$ 0.0052	0.1865 $\pm$ 0.0351	0.2239 $\pm$ 0.0141
— flat MLP on same features	0.1105 $\pm$ 0.0042	0.2117 $\pm$ 0.0075	0.2771 $\pm$ 0.0098	0.1858 $\pm$ 0.0177	0.2253 $\pm$ 0.0062
sampled profile, M=32	0.1286 $\pm$ 0.0075	0.1857 $\pm$ 0.0334	0.2513 $\pm$ 0.0224	0.1802 $\pm$ 0.0032	0.2058 $\pm$ 0.0111
sampled profile, M=64	0.1160 $\pm$ 0.0116	0.1946 $\pm$ 0.0158	0.2467 $\pm$ 0.0135	0.1635 $\pm$ 0.0080	0.2030 $\pm$ 0.0110
<i>conditioning removed at evaluation (symbol-aligned encoder, coefficients kept correct)</i>					
true profile	0.1202 $\pm$ 0.0044	0.1817 $\pm$ 0.0144	0.2619 $\pm$ 0.0158	0.1704 $\pm$ 0.0123	0.2049 $\pm$ 0.0072
shuffled profile	0.3340 $\pm$ 0.0044	0.2958 $\pm$ 0.0078	0.3771 $\pm$ 0.0354	0.2160 $\pm$ 0.0027	0.3195 $\pm$ 0.0131
constant profile	0.3510 $\pm$ 0.0027	0.3246 $\pm$ 0.0170	0.4514 $\pm$ 0.0315	0.1714 $\pm$ 0.0176	0.3241 $\pm$ 0.0204
frozen to $1+\sin(x)$	0.3376 $\pm$ 0.0063	0.3545 $\pm$ 0.0070	0.4303 $\pm$ 0.0113	0.3836 $\pm$ 0.0049	0.3877 $\pm$ 0.0082

Table 15: Latent diagnostics of the specification encoder. R^2 is that of a ridge probe fitted on the *trained* specifications and evaluated on held-out ones; Spearman ρ correlates held-out error with distance to the nearest trained specification, measured in the encoder’s latent and, for comparison, between the raw profiles.

diagnostic	sampled-profile control	symbol-aligned encoder
latent variance explained by the initial condition (%)	62.5	82.4
latent variance explained by the coefficients (%)	11.1	8.4
probe R^2 , held-out IC parameters	0.543	0.615
probe R^2 , held-out family G	0.372	0.664
probe R^2 , held-out family M	0.229	-0.108
ρ (error, latent distance)	0.194	0.335
ρ (error, profile distance)	0.253	0.422

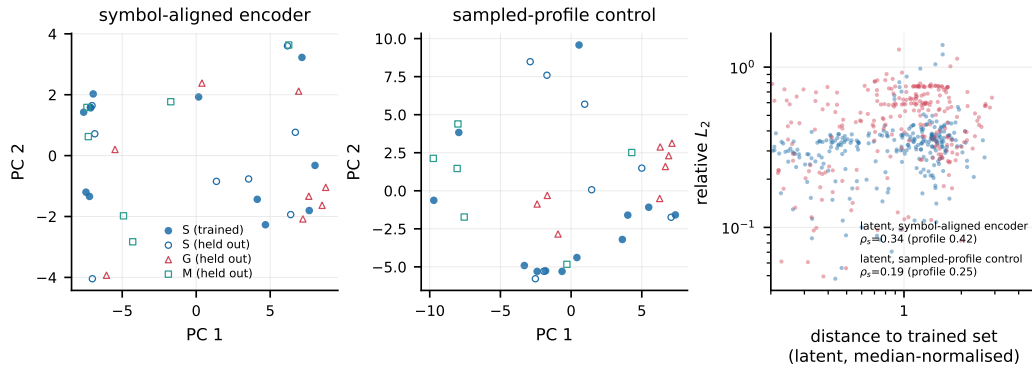


Figure 16: Latent diagnostics on the reaction family, the counterpart of Figure 8. The probe advantage for the structured code on the held-out Gaussian family is the largest of the three families (0.664 against 0.372), and it still does not translate into lower solution error once the denser-sensor control is taken into account.

Table 16: Latent diagnostics of the specification encoder. R^2 is that of a ridge probe fitted on the *trained* specifications and evaluated on held-out ones; Spearman ρ correlates held-out error with distance to the nearest trained specification, measured in the encoder’s latent and, for comparison, between the raw profiles.

diagnostic	sampled-profile control	symbol-aligned encoder
latent variance explained by the initial condition (%)	84.6	79.5
latent variance explained by the coefficients (%)	5.4	12.3
probe R^2 , held-out IC parameters	0.613	0.709
probe R^2 , held-out family G	0.256	0.643
probe R^2 , held-out family M	0.139	0.259
ρ (error, latent distance)	0.351	0.393
ρ (error, profile distance)	0.521	0.413

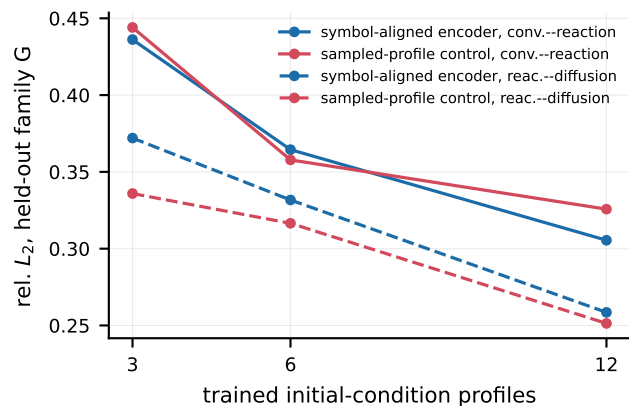


Figure 17: Cutting the number of trained initial-condition profiles from 12 to 6 to 3 does not open a gap between the two representations: whatever the structured encoding contributes, it is not a data-efficiency prior on these families.

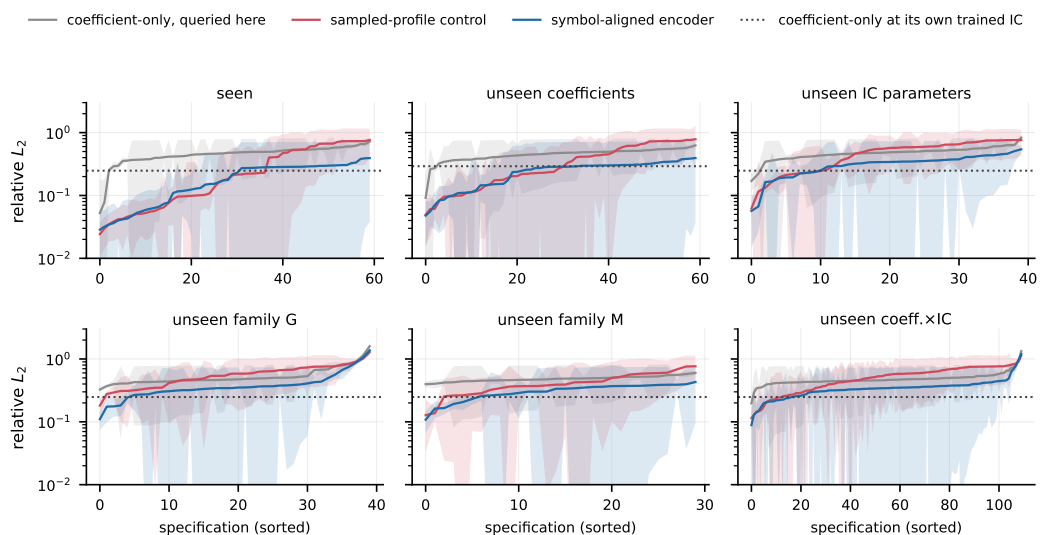


Figure 18: Per-specification error curves on the reaction family, the one family where the two representations separate.